Small Stores, Big Obstacles: Understanding Constraints and Opportunities for Micro-retail Firms

by

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ABSTRACT

Micro and small enterprises (MSEs), particularly informal micro-retailers known as nanostores, play a vital role in developing economies but remain largely underserved by traditional financial institutions and overlooked in economic policy. In Mexico, nanostores account for more than 95% of businesses and over 10% of national employment, yet face high closure rates, low productivity, and limited access to formal credit. This thesis asks: What structural and contextual factors determine the survival and performance of nanostores, and how can policy better support high-potential firms within this segment? To answer this, the study constructs a longitudinal panel of nanostores using microdata from the Mexican Economic Census (2009, 2014, and 2019), and combines it with municipality-level contextual data including crime, infrastructure, unemployment, electricity costs, and business regulations. It applies survival models to estimate firm closure dynamics and implements a misallocation framework to quantify distortions in capital and labor usage. The results reveal that misallocation—particularly of capital—is pervasive and systematically linked to institutional weaknesses and credit access constraints. In response to the limited real-time data available for this sector, the thesis proposes the LIFT Performance Index, developed by the MIT Low-Income Firms Transformation Lab (MIT LIFT Lab), as a diffusion-based tool for monitoring micro-retailers' business sentiments using structured operational surveys. A pilot implementation in Argentina demonstrates the index's potential to generate timely and actionable insights for policymakers and private stakeholders. Overall, this work contributes a novel empirical foundation for understanding heterogeneity within the micro-retail sector and offers a scalable framework for designing targeted, data-driven interventions to support inclusive economic development.

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Chapter 1

Introduction

1.1 Microbusinesses

Over the past two decades, researchers across multiple fields increasingly explore the pressing societal question of why some countries exhibit higher rates of economic growth than others, and how low-performing countries might foster robust economic development. One common approach researchers use to tackle this question examines the firms that produce goods and services within an economy. Firms of all sizes provide employment and income for the majority of households and produce goods and services that constitute a large share of household consumption. As such, firm- and trade-related policies can have far-reaching impacts on poverty reduction, economic growth, and innovation. In some contexts, particularly where the state possesses strong institutional and planning capacity, industrial policy has been associated with improvements in firm performance, innovation, and income growth as evidenced in China. While the Chinese case is unique in many respects, it illustrates how coordinated policy efforts can overcome market failures and accelerate firm development [1, 2].

Micro and small enterprises (MSEs), typically defined as firms with fewer than 50 employees, with microenterprises employing fewer than 10, form the backbone of most economies in terms of firm count and employment [3]. Still, firm-level research has tended to focus on medium- and large-scale formal enterprises, both because of their measurable contributions to aggregate productivity and because they are more visible in administrative and survey data [4, 5]. In contrast, MSEs, despite their prevalence, have only recently gained attention as better data and new methodologies have allowed for more granular analysis [6].

According to the U.S. Small Business Administration, there were approximately 32.2 million small businesses in the United States in 2014, accounting for 43.4% of GDP and 99.9% of all firms [7]. Similar patterns are observed in developing economies: in Mexico, for instance, microbusinesses (firms with less than 10 employees) represent approximately 95.5% of all enterprises, and when combined with small firms (fewer than 50 employees), they account for 99.8% of the business landscape [8]. Comparable figures are found in India, Indonesia, and Nigeria, where most firms employ fewer than 10 workers [9, 10].

The prominence of MSEs raises several unresolved questions. First, why is the distribution of firm sizes so heavily skewed toward small, often unproductive firms? Second, what prevents these firms from growing? Third, if markets function as in neoclassical competitive models, why are not unproductive small firms driven out by competition? Researchers propose several explanations for these patterns. Banerjee and Duflo [11] highlight financial frictions and limited access to credit as key barriers to firm growth in developing countries. Bloom et al. [12] demonstrate how weak managerial practices constrain productivity, particularly among smaller firms. Hallward-Driemeier and Pritchett [13] and La Porta and Shleifer [5] emphasize the role of institutional weaknesses and regulatory burdens. Finally, Restuccia and Rogerson [14] and Hsieh and Klenow [15] show that factor misallocation across firms—caused by distorted markets or poor enforcement—limits aggregate productivity. Understanding the dynamics of firm survival and growth can help inform targeted policies, such as business and management training, access to credit, infrastructure improvements, and better regulatory environments, that improve productivity and economic inclusion.

MSEs face challenges in emerging economies. Because many individuals depend on small businesses for income and employment, strengthening their viability can yield widespread welfare benefits. Ayyagari, Demirgüç-Kunt, and Maksimovic [16] found that firms with fewer than 20 employees generate the largest share of new jobs and often exhibit higher sales and employment growth than larger firms. Identifying what helps small firms survive and thrive could facilitate the emergence of high-growth microenterprises that contribute significantly to job creation [17].

Notably, MSEs in developing countries show lower survival and growth rates than their counterparts in advanced economies. For example, while about one in five new U.S. businesses closes within its first year, the figure is closer to one in three in Mexico [18]. Moreover, transitioning from micro to medium or large size is rare in emerging markets. Whereas nearly 20% of the largest U.S. firms in 2022 originated as SMEs founded after 2000,[1] the equivalent figure is closer to 10% in Mexico, Brazil, and India [19].

[1] Defined as firms with market capitalizations over $10 billion dollars.

1.2 Micro-Retail Businesses

In summary, two main considerations motivate the present research: (1) the sheer prevalence of MSEs in emerging markets given their critical role in income and employment generation, and (2) their relatively poor performance within these settings. Particularly, for this research, we focus on micro-retail businesses. Worldwide there are estimations that there exist more than 50 million micro-retail businesses and this number is projected to continue increasing [20]. These denominated *nanostores* can be found in densely populated urban areas of developing economies, characterized by family ownership and minimal staffing. These establishments create jobs via self-employment and through hiring additional workers. Also, nanostores account for 45% to 95% of the sales channel of consumer packed goods in developing countries [20]. Particularly serving base-of-the-pyramid customers who are often missed by larger retailers.

One question that often arises when studying these nanostores, which the vast majority are informal and seen as unproductive, is why this kind of businesses still exist when larger and better managed firms such as Amazon and Walmart could potentially serve the same market, forcing out nanostores. La Porta and Shleifer [5] offer three views:

1. **Romantic View** (DeSoto's view): Informal firms are seen as entrepreneurial goldmines held back by government barriers. This view sees that informal businesses are creative and potentially productive but are hindered by burdensome regulations, corruption, and bureaucracy. If freed from these constraints through, for example, simpler registration or stronger property rights, they could thrive in the formal economy and fuel national growth.

2. **Parasite View** (McKinsey's View): Informal firms exploit the system and pose a threat to formal businesses. This view, advanced by reports like those from the McKinsey Global Institute, sees informal firms as unfair competitors that avoid taxes and regulations, undermining public services and weakening the incentives for formality.

3. **Dual Economy View** (Lewis/Harris-Todaro Tradition): Informality is a byproduct of poverty and underdevelopment. In this view, formal and informal sectors are fundamentally different. Formal firms are larger, more productive, and run by educated entrepreneurs, while informal firms are small, stagnant, and managed by unskilled individuals producing low-quality goods for poor customers.

This study does not assume any of the three perspectives on informality to be universally valid. While early literature often treats the informal sector as a homogeneous and stagnant segment of the economy, recent advances in firm-level data collection and processing enable more nuanced analysis. These tools make it possible to identify MSEs that are constrained by limited access to capital, labor, or infrastructure rather than inherently unproductive. Using census microdata and original surveys, this research finds evidence that some nanostores exhibit growth potential and actively seek expansion. Accordingly, this study adopts the premise that productive and well-performing businesses can benefit from targeted interventions. By identifying the key constraints that hinder their survival and growth, policymakers and private sector actors can design more effective strategies to support high-potential firms within the micro-retail sector.

The considerable contribution of the micro-retail sector to Mexico's economic activity—both in terms of value added and employment—underscores its relevance as a focus of research. With over one million establishments supporting more than 6 million jobs (around 10% of the total labor force) and contributing roughly 13% of national GDP, the sector is not only economically significant but also deeply embedded in local communities and livelihoods. Its steady growth trajectory over the past two decades, despite major global shocks such as the 2008 financial crisis and the COVID-19 pandemic, further highlights its resilience and structural importance. (See Figure 1.1) Given its scale, economic weight, and social relevance, the Mexican micro-retail sector presents a unique and underutilized opportunity to study the broader microenterprise landscape. This work aims to identify the key factors that enable or hinder the survival and performance of these businesses. By doing so, it seeks to inform targeted interventions —by policymakers, consumer packaged goods firms (CPGs), investors, and development practitioners— that can foster inclusive growth, improve productivity, and support the upward mobility of high-potential microenterprises.

This thesis demonstrates that nanostores, while often informal and constrained, exhibit significant heterogeneity in performance. By identifying the external and internal factors that hinder or enable their survival and productivity, we can inform better-targeted policies and business strategies.

Specifically, the study seeks to answer three main research questions:

- What are the key drivers of nanostore survival, value-added contribution, and marginal productivity?
- How do local contextual variables—such as crime, utility prices, unemployment, and regulations—affect business performance?
- What characteristics distinguish high-performing nanostores from stagnant ones, and to what extent do these characteristics contradict the view that informal microenterprises are inherently unproductive and destined to remain small?

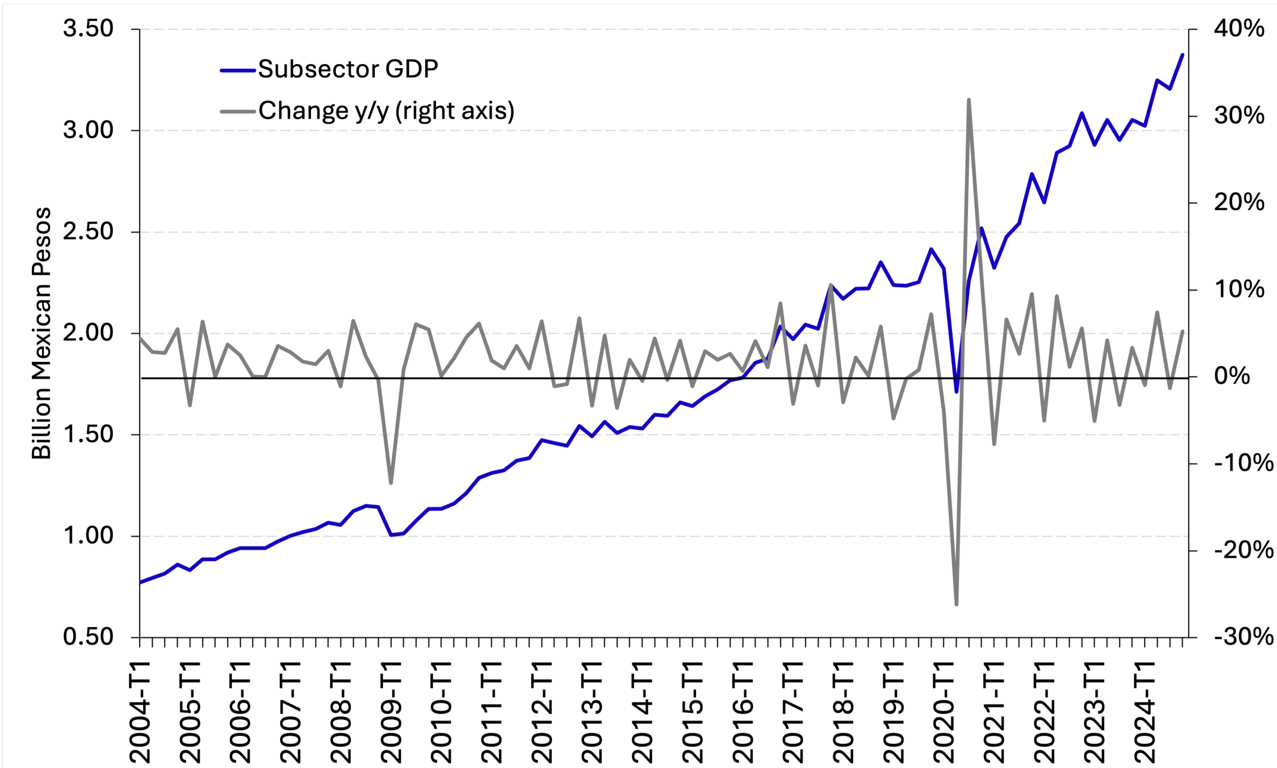

Figure 1.1: Quarterly GDP for the Subsector of Retail Trade of Groceries and Food.

This work contributes to the literature in several ways. First, while previous studies have examined firm survival and productivity in experimental settings or with formal-sector data, this thesis leverages rich census data linked with external indicators —such as crime rates, electricity costs, and institutional quality— to study micro-retail performance at scale in a developing economy. Second, it introduces a firm-level analysis of resource misallocation and productivity dispersion that helps explain why some nanostores remain small while others manage to grow. Third, it offers a novel segmentation of nanostores by performance potential, providing actionable insights for targeted policy and private-sector interventions.

The conceptual framework underpinning this research assumes that nanostore performance is shaped by both firm-level capabilities and external environmental conditions. For instance, crime may raise uncertainty and reduce investment; high electricity prices may depress profit margins; and poor infrastructure may limit access to suppliers and customers. These factors, in turn, influence survival probabilities, productivity, and value-added. By empirically disentangling these pathways, this study aims to clarify which constraints matter most—and for which types of businesses.

Each of the outcome variables is defined with precision in the analysis. The analysis measures survival as the duration of firm activity across census waves. The study infers value-added from reported sales and costs of the goods purchased for resale, controlling for sectoral differences. The model estimates marginal productivity using standard production function approaches, allowing us to infer capital and labor efficiency at the firm level.

This research holds strong policy relevance for public institutions interested in inclusive growth, as well as for consumer packaged goods firms, investors, and development organizations targeting informal retail markets. Rather than advocating for blanket formalization, the findings support nuanced, evidence-based strategies to support high-potential firms within the micro-retail sector.

The thesis proceeds as follows. First, we review the literature on MSE performance, focusing on experimental, theoretical, and other empirical based studies. Second, we outline the various datasets employed, including census data and other external sources that capture local conditions. Third, we estimate the determinants of business survival using Cox proportional hazards and accelerated failure time models. Fourth, we explore the role of input misallocation in constraining productivity, drawing on structural models of firm performance. Finally, we present policy recommendations based on our findings and conclude with reflections on the future of micro-retail in emerging economies.

Chapter 2

Literature Review

In this chapter, we review the literature on the constraints that limit the growth and survival of micro and small enterprises (MSEs), with a particular focus on those operating in developing economies. While supporting MSEs may not drive large-scale productivity gains or structural transformation, these firms play a critical role in poverty alleviation and self-employment, particularly among low-income populations. We aim to understand what prevents these businesses from expanding and which interventions or ecosystem changes may improve their performance.

As described in the introduction, even when our research is focused on the micro-retail sector, particularly in the traditional channel of food and beverages, many research aim at finding the factors affecting small businesses uses observations from retail, manufacturing and/or agricultural sectors. Where relevant, we highlight findings that apply uniquely to retail, given the distinct market structure and operating environment. Throughout the chapter, we use the terms “micro-retail businesses” and “nanostores” interchangeably to refer to micro (less than 10 workers), often informal, retail establishments serving local markets.

The chapter is organized thematically around six key areas identified in the literature: i) human capital and managerial capacity, ii) financial constraints and access to credit, iii) technology adoption and digital infrastructure, iv) institutional and regulatory barriers, and v) the misallocation of capital and labor.

2.1 Human Capital

One frequently cited explanation for the limited performance of small enterprises is inadequate managerial capacity or “business practices.” While it has long been assumed that firms in developing countries are poorly managed, rigorous empirical research on the exact degree of deficiency was limited until Bloom and Van Reenen [21] introduced systematic double-blind interviews to measure management practices. Subsequent studies confirm that variations in management quality can account for significant discrepancies in firm performance both within and across countries [22, 23].

In MSEs, improvements in business practices (e.g., record-keeping, customer relations, basic marketing) can generate measurable gains in profits and productivity [24]. However, the literature often notes methodological challenges such as limited sample sizes, short time horizons, and substantial attrition—problems exacerbated by the high failure rates of microenterprises. Additionally, although business training programs have led to improved managerial practices, the effect on long-term profits remains inconsistent, potentially because of limited statistical power, non-random attrition, or external market conditions [25, 26].

Formal education and entrepreneurial experience may also contribute to better outcomes in small businesses. Although some scholars posit a positive relationship between education and firm performance, others find mixed results, particularly for smaller firms with fewer paid employees [27–29]. Experience offers hands-on skills development, yet the extent of its impact may hinge on sector-specific factors and local market characteristics [30].

An important distinction in understanding the performance of MSEs lies in the motivation behind business formation. The literature differentiates between "necessity entrepreneurs," who operate businesses due to the absence of viable wage employment opportunities, and "opportunity entrepreneurs," who create firms to exploit perceived market opportunities [31]. This distinction has direct implications for policy: while both groups may benefit from improved access to finance or training, the returns to such interventions are often concentrated among the latter.

Necessity entrepreneurs may lack the intrinsic motivation, skills, or long-term orientation needed to scale their businesses, using self-employment instead as a coping mechanism for economic insecurity [32]. In contrast, opportunity-driven entrepreneurs tend to invest more, innovate, and hire outside labor at higher rates. As a result, blanket interventions that assume all microenterprises seek growth may have muted average effects if a large share of firms are operating solely to survive. Empirical work has shown that targeting high-aspiration or growth-oriented firms may be a more effective strategy for achieving productivity gains [6].

2.2 Financial Constraints

Financial constraints constitute a critical barrier limiting the growth and performance of MSEs, especially in emerging economies. Access to affordable, reliable finance is often cited as one of the primary factors influencing the survival, scalability, and productivity of small firms [33, 34]. MSEs typically operate with limited collateral, uncertain cash flows, and minimal credit histories, making them particularly unattractive to formal financial institutions. Consequently, these enterprises frequently rely on informal financial mechanisms or self-financing, which can constrain their ability to invest, innovate, and expand [35].

The existing literature underscores that small firms in developing contexts commonly face severe credit rationing, driven primarily by information asymmetries between lenders and borrowers [36]. Formal lenders often lack the mechanisms to accurately assess the creditworthiness of business owners or perceive high transaction costs relative to loan sizes. As a result, microentrepreneurs resort to alternative sources of credit such as suppliers, informal money lenders, cooperatives, and family networks, which may offer easier access but frequently come with higher costs and limited availability [37, 38].

Empirical research suggests that relaxing these financial constraints can significantly improve enterprise outcomes. For example, studies through randomized controlled trials have documented positive returns to capital injections in MSEs across several developing countries. Mel, McKenzie, and Woodruff [34] showed that providing small grants or low-cost loans to microentrepreneurs in Sri Lanka substantially increased profits, highlighting the productive potential constrained by limited financing. Similarly, Banerjee and Duflo [11] illustrate how increased access to formal credit through microfinance programs positively impacts firm growth and productivity, although effects vary depending on loan conditions and firm characteristics.

Additionally, the availability of supplier credit, or trade credit, has emerged as an important channel for alleviating financial constraints among MSEs. Supplier credit enables MSEs to manage working capital more effectively and smooth consumption fluctuations, thereby indirectly enhancing productivity and growth [39]. However, dependence on trade credit alone can limit the capacity of businesses to scale and diversify, underscoring the importance of integrating formal financial solutions tailored to MSEs needs.

In summary, addressing financial constraints through improved access to credit, reduction of information asymmetries, and promotion of inclusive financial products could substantially enhance microenterprise performance. Policymakers and financial institutions must thus focus on creating conducive environments that bridge the gap between MSEs and formal financial systems.

2.3 Technology Access

Technology access represents another critical factor influencing the performance of MSEs, particularly in developing economies. These firms frequently lag behind larger firms in adopting new technologies due to limited financial resources, lack of awareness, and inadequate infrastructure [4]. These constraints often result in a technology gap, limiting productivity, efficiency, and competitiveness.

Recent studies emphasize the transformative potential of technology adoption among MSEs. Simple technologies, such as mobile payment platforms, digital record-keeping, and online marketplaces, can significantly improve operational efficiency, market reach, and financial management [40, 41]. For instance, mobile money services have shown considerable impacts on reducing transaction costs, enhancing financial management, and facilitating access to broader markets and suppliers [42].

Despite these benefits, barriers to technology adoption persist. High initial costs, insufficient digital literacy, unreliable electricity and internet access, and resistance to change frequently hinder technological upgrading [43]. Furthermore, information asymmetries regarding available technologies and their benefits exacerbate slow diffusion rates among smaller enterprises.

Empirical evidence highlights the positive impacts of targeted interventions designed to improve technology access. Training programs focused on digital literacy, subsidies for technological equipment, and infrastructure development initiatives have demonstrated measurable gains in firm productivity and growth [44]. Additionally, policy-driven initiatives promoting affordable internet and electricity infrastructure have enabled broader technology adoption, thereby enhancing competitiveness and market participation among MSEs [45].

In conclusion, improving technology access through targeted support, infrastructure development, and digital literacy training can substantially enhance the productivity and growth of microenterprises. Policymakers and stakeholders must prioritize initiatives that bridge technological gaps to facilitate sustainable economic development.

2.4 Institutional Barriers

Institutional barriers, including regulatory frameworks, legal systems, and crime, significantly influence the operational environment and performance of MSEs. These barriers can disproportionately affect small firms, impeding their ability to grow, invest, and remain competitive [46].

Excessive regulations and cumbersome administrative processes can create substantial entry barriers and operational burdens for MSEs. Empirical studies indicate that heavy regulatory burdens and inefficient bureaucracies frequently lead to informality among small firms, limiting their access to formal credit, legal protections, and public resources [5]. Additionally, unclear or unpredictable legal systems exacerbate uncertainty, deterring investment and discouraging firm growth [47].

Crime and violence represent additional critical challenges. High levels of crime impose direct costs through theft and property damage, and indirect costs due to increased security expenditures and disruptions to normal business operations [48]. MSEs, often operating with thin margins, are particularly vulnerable to such disruptions. Studies from Latin America and Africa document that crime significantly reduces firm profitability, discourages investment, and adversely affects business sustainability [49].

Addressing institutional barriers requires comprehensive policy interventions. Simplifying regulatory procedures, enhancing transparency and predictability in the legal environment, and strengthening public safety measures can markedly improve businesses viability [50]. Policymakers should prioritize reforms aimed at reducing administrative burdens and enhancing legal certainty, alongside targeted crime reduction programs, to foster a more conducive ecosystem for MSEs growth.

2.5 Resource Misallocation

Beyond firm-level deficiencies and external barriers, structural misallocation of resources presents a fundamental constraint to MSEs growth and aggregate productivity. In theory, efficient markets should allocate inputs such as capital and labor to the most productive firms [51]. However, a growing body of literature demonstrates that in many developing countries, this process breaks down, with more productive firms often receiving disproportionately fewer resources than their less productive counterparts [14, 15].

Misallocation can arise from a variety of frictions: credit constraints, labor market segmentation, informality, and discriminatory regulations. These distortions prevent factor inputs from flowing to their most efficient uses, leading to both lower firm-level output and substantial aggregate productivity losses [52]. In the context of MSEs, these inefficiencies are magnified due to limited access to formal financial systems, rigidities in labor mobility, and lack of institutional support. As a result, firms that could otherwise expand and contribute meaningfully to employment and value-added remain artificially small or stagnant.

Empirical studies using firm-level census data from countries such as India, China, and Mexico find that correcting these distortions could lead to productivity gains ranging from 30% to over 50% [15, 53]. In the Mexican context specifically, significant dispersion in marginal products of capital and labor among MSEs suggests persistent inefficiencies in resource allocation [54]. Importantly, recent work by Lagakos [55] highlights that misallocation of labor across heterogeneous firms can explain a substantial portion of cross-country income differences. They find that low-productivity firms in low-income countries are often too large relative to their efficiency, while high-productivity firms are too small—indicative of deep structural barriers that prevent efficient reallocation of workers.

Understanding misallocation is therefore critical for designing effective policies. Interventions that reduce frictions in input markets—such as improving credit access, simplifying labor contracts, or formalizing property rights—can have multiplicative effects by enabling more productive firms to expand while crowding out persistently unproductive ones. Importantly, such reforms complement rather than substitute micro-level interventions, reinforcing the need for a dual approach that addresses both firm capabilities and structural constraints.

In sum, resource misallocation constitutes a central, yet often underappreciated, mechanism limiting the growth and productivity of MSEs. Addressing this problem requires a systemic view of firm dynamics, emphasizing not just the average firm, but the misaligned allocation of resources across the entire distribution of businesses.

2.6 Market Access and Competition

Limited access to product markets constitutes another major constraint on MSEs growth, particularly in the retail sector. Geographic isolation, weak transportation infrastructure, and fragmented supply chains often raise input costs and reduce potential customer bases for small businesses [56]. In such settings, even productive firms may fail to scale if they cannot connect to broader or more profitable markets.

MSEs also operate in highly localized markets with intense competition and thin margins, which may discourage innovation or expansion. Unlike larger firms that can absorb fixed costs or cross-subsidize across locations, nanostores often lack the scale economies needed to compete with supermarkets or e-commerce platforms. In addition, low customer mobility in informal settlements and limited consumer purchasing power restrict growth opportunities [41].

Empirical work highlights that infrastructure improvements—such as better roads, digital connectivity, or logistics support—can significantly increase market access and improve firm outcomes [57]. Interventions that reduce spatial and logistical frictions thus hold promise for expanding the effective market radius of MSEs.

2.7 Literature Gap

The literature reviewed underscores several persistent constraints faced by MSEs in developing countries—including limited managerial capacity, financial exclusion, institutional barriers, technology gaps, and substantial resource misallocation. While extensive research has explored firm-level interventions through randomized evaluations or modeled macroeconomic distortions across sectors like manufacturing and agriculture, critical gaps remain in understanding the specific dynamics of survival and productivity in the micro-retail sector, particularly in low- and middle-income countries.

Existing studies often suffer from small sample sizes, lack longitudinal depth, or fail to distinguish between stagnant businesses and those that are high-potential but constrained. Moreover, the phenomenon of misallocation has been extensively studied in formal production sectors but remains understudied in informal retail contexts, where firms operate with minimal records and in highly localized, fragmented markets.

This thesis seeks to fill these gaps by leveraging Mexican national economic census data that enables longitudinal tracking of nanostore survival and productivity. It incorporates a range of contextual variables—such as crime, infrastructure quality, unemployment, and credit access—motivated by existing literature, but rarely combined at scale. While prior studies have focused on forecasting performance or identifying high-growth firms ex ante [6, 58, 59], this research takes a complementary approach: rather than building predictive algorithms, it aims to understand and compare the structural and contextual factors that shape performance, with the goal of informing more targeted and efficient policy interventions.

Methodologically, the thesis also contributes by constructing a real-time, index-based framework to monitor micro-retail performance—providing a scalable and policy-relevant tool to identify trends, inform resource allocation, and ultimately support inclusive economic development. In doing so, it offers new insights into which firms survive, which thrive, and, critically, under what conditions.

Chapter 3

Data Description

This chapter describes the main datasets used in the analysis, with a primary focus on the firm-level data provided by Mexico's Economic Census. We begin by detailing the scope, structure, and content of the census, particularly its relevance for studying micro-retail businesses over time. We then describe the process of constructing a panel of nanostores across three waves (2009, 2014, and 2019) and highlight key trends in their characteristics, size, and geographical distribution. Finally, we present the set of contextual variables, covering crime, unemployment, utility prices, population density, and business environment indicators, that are merged to the census data at the municipality or neighborhood level. Together, these datasets provide a rich foundation to analyze survival, growth, and productivity among nanostores in Mexico.

3.1 INEGI Census Data

In this study, we use data from the Mexican Economic Census conducted by the Instituto Nacional de Estadística y Geografía (INEGI), the national agency responsible for statistical and geographical information in Mexico. Since 1930, these censuses have been conducted every five years, originally focusing on the manufacturing sector but progressively expanding to include trade, services, transportation, construction, and other non-agricultural activities. The Economic Census is one of the most comprehensive and granular statistical efforts in Latin America and serves as the principal data source for understanding the structure and dynamics of Mexico's business sector.

The census aims to provide a full accounting of all economic units operating in the country, with the exception of agriculture, forestry, and fishing (which are covered by a separate census). In its most recent iterations (2009, 2014, and 2019), the census has employed a dual strategy combining exhaustive enumeration for small establishments and sample-based collection for medium and large firms. Microenterprises and informal establishments—especially those operating in trade and services—are typically enumerated exhaustively, which makes the census uniquely valuable for studying the informal and micro-retail segments.

Data collection involves in-person interviews by trained enumerators, using standardized questionnaires tailored to each sector. For informal or non-registered firms, enumerators are trained to identify businesses operating without legal registration or fixed locations. The census captures detailed firm-level information, including total revenues, costs, employment, wages, inventories, investment, credit access, ownership, and basic digital technology usage. It also includes georeferenced location data at the block level, enabling spatial analysis of business dynamics.

One notable strength of the INEGI Economic Census is its temporal consistency and comparability across waves, which allows this work to construct longitudinal panels by tracking businesses that remain in operation across census rounds. INEGI assigns a unique business identifier, which permits matching establishments over time. However, not all establishments can be tracked longitudinally due to firm closures, relocations, or changes in reporting practices. Attrition and ID inconsistencies are particularly common among informal or home-based businesses.

Despite these challenges, the census remains one of the few sources in the developing world that provides near-complete coverage of microenterprises at national scale, including firms with fewer than five employees that are often excluded from enterprise surveys. This level of granularity is particularly important for the present study, which focuses on nanostores and other small retailers. The 2019 wave, for instance, recorded over 4.9 million economic units nationwide, more than 95% of which had fewer than 10 employees.

Our analysis is focused on nanostores. Particularly, this sector encompasses food, beverages, and tobbaco retailers which is represented by the cod 461110 of the North American Industry Classification System (NAICS). Retail trade is of particular interest as it represents roughly 20% of total employment in Mexico, and micro- and small enterprises as a whole employ approximately 60% of the workforce [60]. Moreover, micro-retail businesses provide critical points of access for local consumers, with over 70 million people relying on these shops for groceries and other daily necessities. Such shops also account for around 50% of consumer-goods sales by large CPG companies (Figure 3.1).

Regarding geographical distribution, the central and southern regions of Mexico contain the largest number of nanostores. Municipalities such as Iztapalapa (Mexico City), Ecatepec (State of Mexico), Puebla (Puebla), León (Guanajuato), and Guadalajara (Jalisco) have notably high concentrations (Figure 3.2). When adjusting for population, Oaxaca, Tlaxcala, and Puebla lead the country in the number of nanostores per 1,000 residents (Figure 3.3).

Figure 3.1: Example of a micro-retail store in Mexico City.

Beginning in 2009, INEGI introduced an establishment identifier, enabling researchers to track the same businesses across multiple census waves and to analyze micro-retail dynamics longitudinally. Although micro-retail establishments vastly outnumber small retail establishments, there is some evidence of upward mobility, which means that stores that started as micro grew till become small (Table 3.1). For example, 261 businesses categorized as microenterprises (less than 10 employees) in 2009 had grown sufficiently to be reclassified as small enterprises by 2014. Understanding the factors that enable such transitions is a key objective of this study.

Observations per wave			
	Total Stores in		
	2009	**2014**	**2019**
Micro	578,277	587,575	573,923
Small	1,768	1,364	218
Total	**580,045**	**588,939**	**574,141**
Stores going from micro to small	-	261	103

Table 3.1: Store Counts Across Census Waves

Microenterprises are defined as establishments with fewer than 10 workers. In our data, almost 50% of nanostores have between 1 and 3 workers. Over the years, the number of workers per store seems to be reducing, which further shows the low growth.(Figure 3.4) Previous

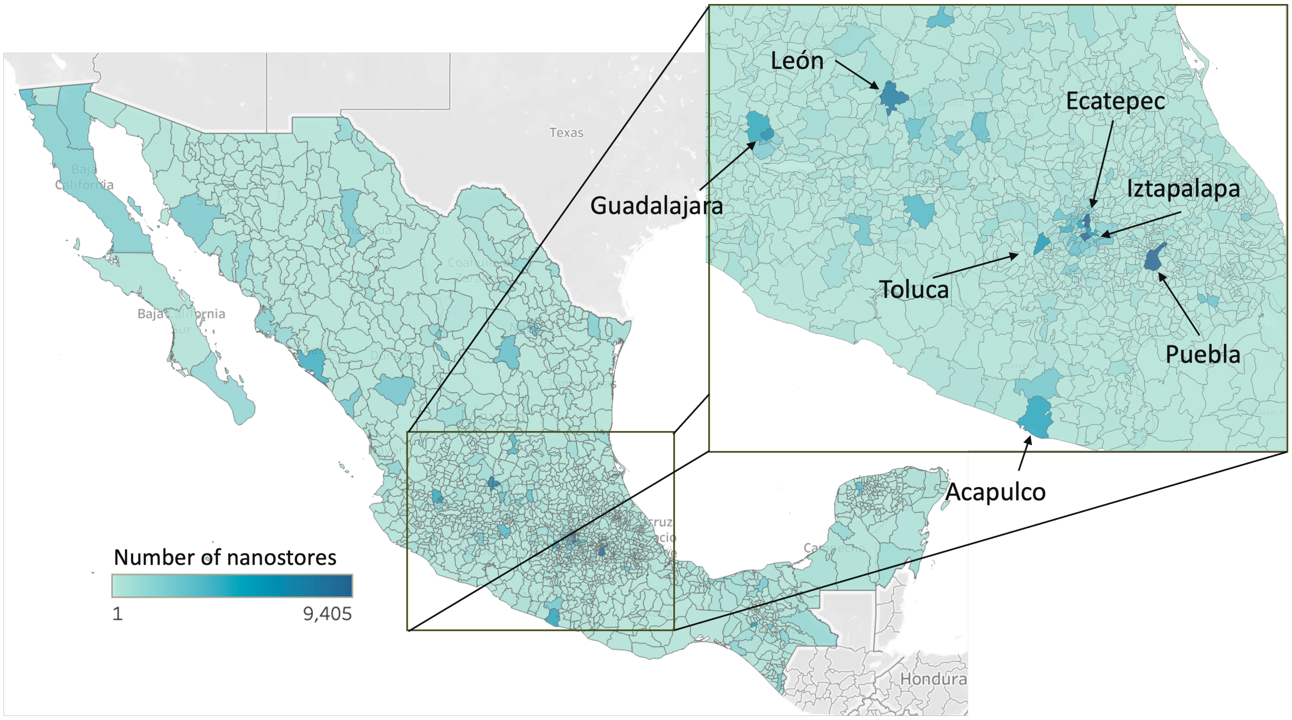

Figure 3.2: Number of nanostores by municipality.

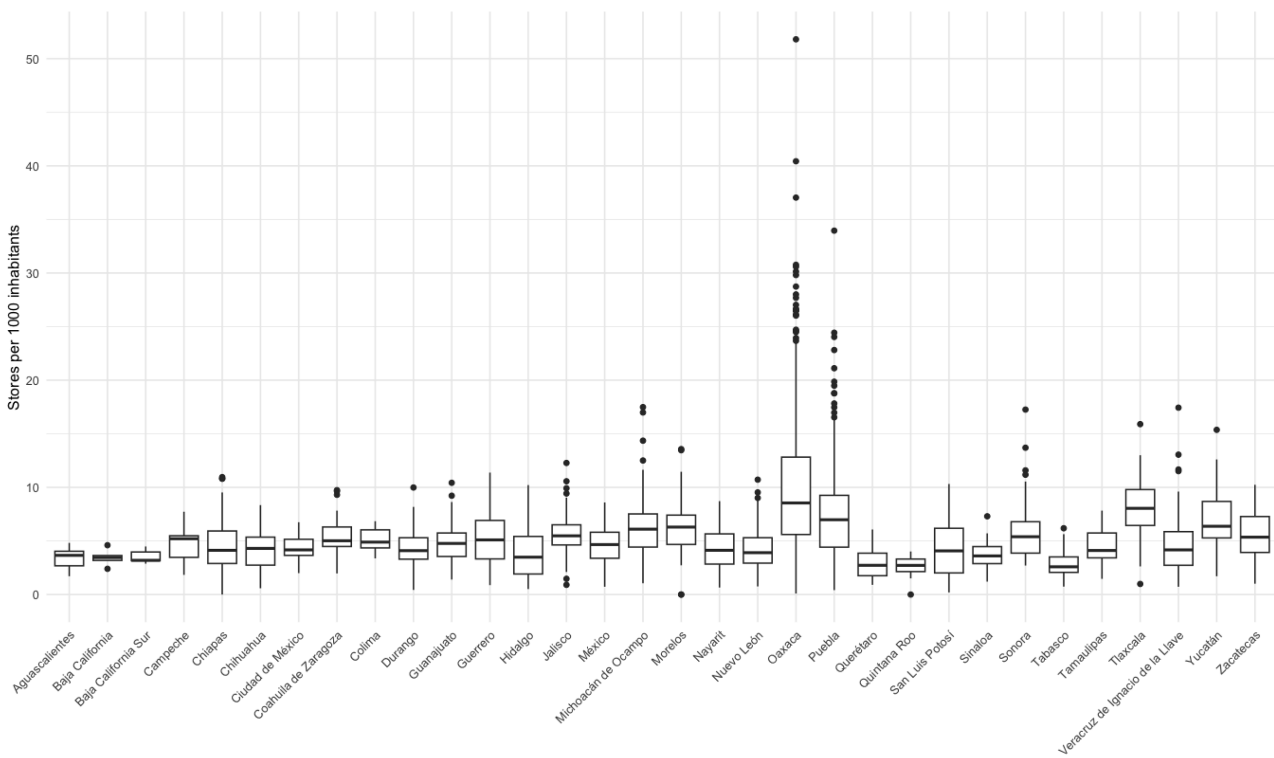

Figure 3.3: Number of nanostores per 1,000 inhabitants, by municipality.

research has identified "gazelles" as firms achieving annual growth in sales (or jobs) of at least 20% from a base above $100,000 [61]. High-growth microbusinesses are particularly interesting due to their potential outsized impact on employment. By identifying the attributes that allow certain firms to "take off," policymakers, investors, and other stakeholders can tailor interventions to foster faster economic development [58, 59].

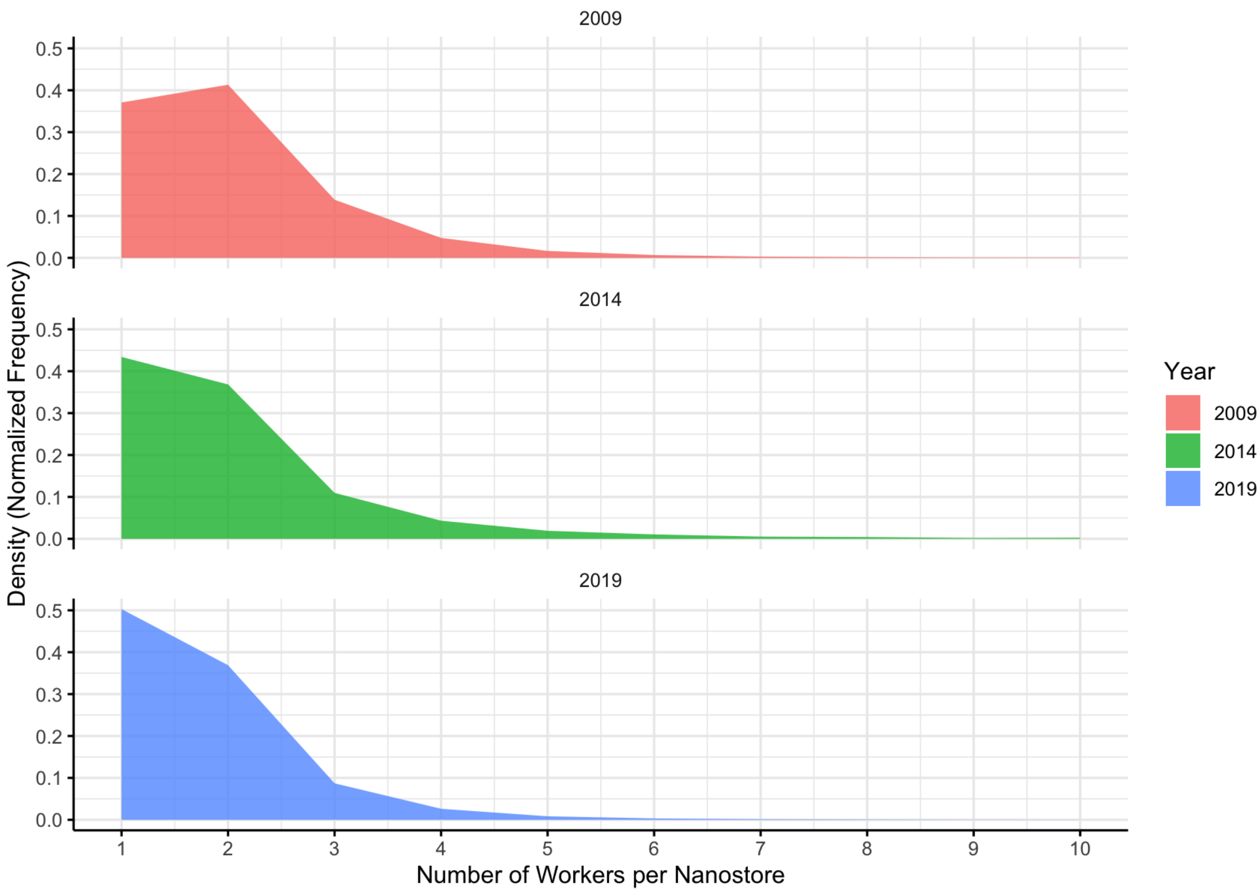

Figure 3.4: Distribution of the number of workers per store.

Although the specific survey questions posed by INEGI have varied across census waves, core inquiries on demographics, finance, credit, employment, and assets have remained relatively consistent. Variables collected refer to business performance in the year preceding each census. In 2019, INEGI also introduced perception-based questions addressing business constraints. Most respondents cited insecurity and high utility costs as their primary concerns, with owners in Quintana Roo reporting particular difficulties related to government regulations, such as bureaucracy and the process of obtaining operating permits (Table 3.2). Interestingly, fewer nanostores reported barriers related to technology access, possibly reflecting limited awareness of how technology adoption could enhance business performance.

As noted by Comin and Hobijn [43], the diffusion of new technologies is often slower in lower-income settings not merely due to financial constraints, but also because of institutional and informational frictions that limit firms' awareness and capacity to adopt available innovations.

	Low Quality of Raw Materials	Lack of Technology Access	Low Experienced Workers	Government Licenses and Permits	Bureaucracy	Corruption	Informal Competition	High Taxes	Lack of Credit	Low Demand	High Competition	High Supplier Costs	High Utilities Prices	Insecurity
National	**0.49**	**0.93**	**0.95**	**3.21**	**3.50**	**4.72**	**8.14**	**8.92**	**9.11**	**16.85**	**21.67**	**22.10**	**29.66**	**34.45**
Aguascalientes	0.93	1.84	1.90	3.59	4.92	4.63	7.88	11.11	11.02	13.92	19.87	21.41	51.00	32.93
Baja California	0.55	1.11	1.54	8.64	9.07	6.50	7.18	12.81	6.06	10.10	14.26	13.49	28.59	50.12
Baja California Sur	0.78	1.56	1.61	8.87	9.37	4.92	5.79	15.48	14.29	11.90	12.26	21.45	51.31	26.60
Campeche	0.56	1.29	0.89	7.92	7.88	3.88	12.55	18.26	12.30	19.92	19.78	19.96	43.64	16.95
Coahuila de Zaragoza	0.40	0.69	0.71	2.33	2.83	2.96	5.03	8.74	7.00	9.61	13.14	13.08	31.70	17.50
Colima	0.67	1.46	0.83	8.75	9.15	5.83	8.02	20.74	15.38	15.61	16.84	22.20	61.09	32.32
Chiapas	0.63	0.86	0.64	1.00	1.14	2.64	8.92	6.93	12.29	20.27	22.39	26.10	20.35	18.14
Chihuahua	0.44	1.15	1.03	3.20	3.15	3.63	5.61	9.90	6.37	10.64	12.72	14.59	37.89	34.21
Ciudad de México	0.72	1.34	1.18	2.61	3.93	7.23	11.37	7.64	9.20	15.44	26.73	22.51	21.70	50.94
Durango	0.37	1.16	0.75	3.12	3.89	3.71	6.35	9.38	10.17	14.75	15.49	24.79	37.85	18.21
Guanajuato	0.45	0.67	0.92	1.84	2.62	3.99	5.96	6.70	6.85	14.89	15.96	14.52	29.22	48.36
Guerrero	0.45	0.85	0.65	1.85	1.74	5.52	7.79	7.02	13.77	16.18	17.67	23.95	33.96	27.18
Hidalgo	0.65	1.62	1.16	5.35	5.17	4.90	12.13	11.30	12.75	24.76	30.42	30.84	32.06	23.25
Jalisco	0.42	0.79	1.05	5.17	4.36	4.07	5.68	11.48	4.93	10.98	14.90	13.99	40.49	33.82
Estado de México	0.42	0.77	0.95	3.10	3.53	6.56	8.97	7.97	6.70	16.46	29.26	21.40	16.86	44.95
Michoacán de Ocampo	0.59	1.03	0.95	2.35	2.18	4.89	7.70	9.11	9.26	18.49	20.08	23.37	34.14	24.41
Morelos	0.59	0.83	1.44	3.46	3.15	8.38	9.33	7.69	9.44	21.83	25.39	22.28	43.24	45.07
Nayarit	0.61	1.33	0.85	5.25	4.84	2.87	8.29	12.67	7.99	16.67	17.40	24.40	54.35	12.71
Nuevo León	0.39	0.66	0.72	1.93	1.77	3.32	4.81	5.46	4.65	8.29	9.99	9.65	27.78	46.09
Oaxaca	0.48	1.00	1.03	1.60	1.97	2.30	8.55	8.09	12.22	24.24	23.17	34.96	30.17	16.13
Puebla	0.56	0.96	0.85	3.06	3.63	4.93	9.42	9.57	9.51	18.99	30.84	29.36	25.96	44.11
Querétaro	0.63	1.59	1.30	11.84	13.90	4.63	10.46	14.81	8.51	14.73	20.18	23.18	40.81	36.95
Quintana Roo	0.86	1.96	2.35	26.84	24.55	15.11	8.49	29.86	10.85	7.48	17.09	14.71	42.66	47.90
San Luis Potosí	0.43	1.14	1.28	4.39	4.81	3.55	7.02	10.12	9.04	14.86	16.02	18.71	36.72	35.40
Sinaloa	0.31	0.41	0.42	0.95	1.08	2.23	4.97	6.89	9.28	15.38	14.74	20.30	35.94	19.30
Sonora	0.41	0.71	0.79	2.04	1.81	3.08	5.45	8.00	6.70	12.32	11.06	20.14	40.36	33.13
Tabasco	0.27	0.33	0.52	0.76	1.11	3.90	4.22	5.31	15.94	20.11	14.26	22.36	16.07	32.74
Tamaulipas	0.28	0.79	1.33	2.51	2.42	4.61	5.33	7.90	9.06	14.49	11.51	15.75	34.52	32.52
Tlaxcala	0.46	0.94	0.94	3.88	4.76	3.89	10.22	10.43	14.58	19.68	32.17	28.07	25.80	28.61
Veracruz de Ignacio de la Lla	0.33	0.54	0.56	1.21	1.44	3.48	8.53	6.21	9.00	22.98	20.75	24.26	27.51	32.57
Yucatán	0.44	1.14	0.88	3.62	4.65	1.84	11.32	12.52	11.21	14.59	28.13	25.24	51.65	7.34
Zacatecas	0.62	1.83	1.14	2.89	3.28	5.50	8.46	11.26	13.51	21.51	18.18	22.94	34.44	32.16

Table 3.2: Challenges faced by nanostore owners.

Overall, the INEGI Economic Census provides an unparalleled lens on Mexico's microenterprise landscape. Our working dataset consists of approximately 1.743 millions unique micro-retail establishments tracked across three census waves (2009, 2014, 2019). The data enable us to measure firm survival, employment, and basic financial indicators while observing spatial and temporal trends at high geographic resolution. To provide an overview of the core characteristics of micro-retail businesses in our sample, Table 3.3 presents descriptive statistics for key firm-level variables across the three most recent census waves. All values reflect nanostores classified under NAICS code 461110 (Food, Beverage, and Tobacco Retailers) and are limited to microenterprises with fewer than 10 employees. This summary highlights patterns, including a gradual decline in average number of workers and nanostores per census tract (each census tract has between 2500 and 8000 people), as well as persistently low rates of credit and bank card usage.

Table 3.3: Descriptive Statistics of Micro-Retail Businesses by Census Wave

	2009		**2014**		**2019**	
Variable	Mean	SD	Mean	SD	Mean	SD
Number of employees	2.83	2.18	1.95	0.58	1.72	0.41
Yearly revenue (Thousand MXN)	238.20	242,781	221.67	283.14	261.47	138.77
Operational assets (Thousand MXN)	57.47	230.45	79.23	367.52	98.13	129.28
Inventory stock level (Thousand MXN)	10.97	40.18	8.79	64.58	16.75	37.38
Has credit (dummy)	0.12	0.33	0.18	0.16	0.13	0.13
Has a bank card (dummy)	0.04	0.8	0.06	0.12	0.03	0.08
Number of nanostores per AGEB	31.01	15.78	30.99	40.90	16.5	12.56
Age of business (years)	7.84	4.05	7.53	3.95	10.29	4.96

3.2 Contextual Data

To better understand the role of local conditions in shaping nanostore survival and performance, we complement the census microdata with a comprehensive set of contextual variables at the municipality and sub-municipality level. These data allow us to capture structural barriers and environmental factors that affect both business owners and consumers. All variables are cross-sectional and refer to 2018 or 2019 to match the timing of the most recent Economic Census wave. Below, we describe each set of variables used.

3.2.1 Institutional Quality and Crime

To study the impact of insecurity and weak rule of law, we use municipality-level crime data from Mexico's federal open data portal [62]. Following the literature on underreporting and measurement error in crime statistics [63], we restrict our analysis to property-related crimes that are more likely to be reported and systematically tracked. Specifically, we aggregate counts of home burglary, auto theft, cargo theft, business robbery, and property damage during 2018. This approach captures both direct threats to nanostores and broader deterrents to commerce.

3.2.2 Labor Market Conditions

We incorporate unemployment data from the Indicadores Laborales para los Municipios de México (ILMM), produced by INEGI through small-area estimation techniques based on the Encuesta Nacional de Ocupación y Empleo (ENOEN).[1] These estimates provide municipality-level rates of total and informal unemployment. Labor market slack can have two opposing effects: it may reduce local purchasing power, lowering demand for goods, or alternatively increase entrepreneurship as unemployed individuals start necessity-driven businesses [65].

3.2.3 Utility Costs

High utility prices are often cited by nanostore owners as a critical business constraint. To operationalize this, we web-scraped the monthly fixed and variable electricity tariffs for low-demand commercial users across municipalities using historical data from Mexico's Federal Electricity Commission [66]. In 2018, fixed prices ranged from 38 to 93 MXN per month depending on region and usage tier. These rates vary due to demand, subsidies, and regional climate factors.

3.2.4 Population and Density

Both nanostore prevalence and crime intensity are likely correlated with local population levels. To control for this, we use INEGI's 2019 population census data at the AGEB (Área Geoestadística Básica) level—the finest census tract available. This enables precise calculation of nanostores per 1,000 inhabitants and allows us to examine whether store density is driven by consumer proximity, as suggested in prior work on spatial frictions in micro-retailing [55].

[1] ILMM follows World Bank small area estimation guidelines [64]

3.2.5 Car Ownership and Consumer Mobility

Following the framework of Lagakos [55], we include a proxy for consumer mobility: the number of private vehicles registered per 1,000 inhabitants in each municipality. Data come from INEGI's Estadística de Vehículos de Motor Registrados en Circulación (VMRC) [67]. Higher car ownership suggests a greater ability to travel to supermarkets or other retail options, which could increase competitive pressure on nanostores and affect their survival or sales.

3.2.6 Local Development: Human Development Index

To capture multidimensional socioeconomic development, we use the Human Development Index (HDI) at the municipal level from the United Nations Development Programme (UNDP). The 2010–2020 Municipal HDI report includes updated metrics on income, education, and health for all municipalities in Mexico [68]. HDI provides a summary measure of overall living standards, which can affect both consumer demand and entrepreneurial capacity.

3.2.7 Human Capital: Education Index

As a proxy for local workforce quality, we construct an education index based on the weighted average educational attainment of nanostore employees, using data from the Economic Census. Workers are classified into four categories: no education, basic, middle, and higher education. Each level is assigned a weight reflecting years of schooling, and the index is computed as a continuous score for each business. This allows us to control for differences in human capital endowment that could affect store performance, innovation, and formality.

3.2.8 Business Environment: Doing Business Index

We focus specifically on the "Apertura de una empresa" score, which measures the regulatory burden of formally creating a new firm in each of Mexico's 32 states. This sub-indicator aggregates four components: Procedures, Time, Cost, and Paid-in Capital.

- **Procedures:** Number of distinct steps required by federal, state, and municipal authorities (e.g. name reservation, tax-ID registration, municipal license, social-security enrollment). Simultaneous requirements count as one procedure, but each separate office visit counts individually.

- **Time:** Total calendar days from submission to receipt of all required approvals, assuming entrepreneurs have perfect information and incur no unofficial delays.
- **Cost:** Sum of all mandatory fees and legally required professional (notarial) charges, expressed as a percentage of per-capita income. No informal payments are counted; expert notary estimates fill gaps where official schedules are unavailable.
- **Paid-in Capital:** Minimum cash or asset deposit required by law (held in escrow or by a notary), also as a share of per-capita income, reflecting upfront liquidity needs just to incorporate.

This metric matters for our study because high entry barriers discourage formalization among small, potentially high-productivity nanostores. States with lower scores may see fewer innovative entrepreneurs enter the formal sector, forcing resources to remain in lower-productivity informal outlets and exacerbating misallocation in micro-retail markets [69].

Taken together, these contextual variables allow us to account for a wide range of environmental conditions that may influence the behavior and outcomes of micro-retail establishments. By linking firm-level outcomes to these locality-specific characteristics, we aim to identify both enabling factors and structural barriers to nanostore growth and survival.

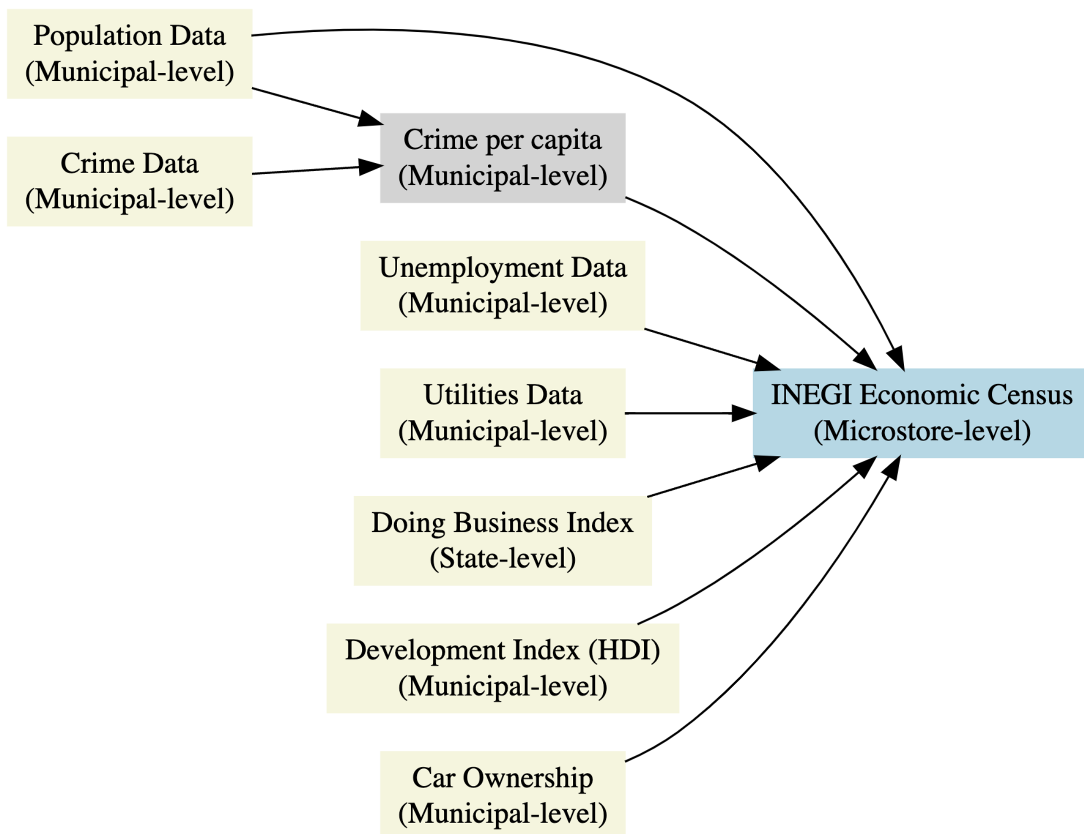

Figure 3.5: Data flowchart showing the level of variation for each variable.

Chapter 4

Survival Rate

Understanding the survival dynamics of nanostores is crucial for identifying barriers to business longevity and designing interventions that can promote firm resilience. Between 2009 and 2014, 59.43% of nanostores observed in 2009 remained operational, meaning that 344,750 out of 580,040 micro and small establishments closed within five years. Between 2014 and 2019, the corresponding five-year survival rate rose slightly to 62.47%. Over a ten-year period from 2009 to 2019, the survival rate declined to approximately 42.15%, highlighting a high rate of attrition among micro-retailers.

Figure 4.1 illustrates firm dynamics across census waves. Over five-year intervals, most nanostores are survivors from the previous wave. In contrast, over ten years, newly established firms constitute a majority of active businesses. Despite this churn, the total stock of nanostores remained relatively stable across the three waves—hovering around 600,000 establishments—suggesting a dynamic equilibrium where firm entry offsets firm exit.

There is significant regional variation in survival outcomes. For instance, from 2014 to 2019, the State of Mexico recorded the highest survival rate at 67%, while Sonora posted the lowest at 49.5% (Figure 4.2). These differences may reflect local market conditions, regulatory environments, and broader development indicators.

To formally estimate survival probabilities over time, we employ a nonparametric Kaplan–Meier survival analysis [70]. This estimator accounts for right-censoring—i.e., firms that remain active at the end of the study period—and models time-to-event based on the year of business founding and observed exit. We assume that businesses not observed in the subsequent census wave closed within one year after their last appearance. Under this assumption, approximately 16.5% of nanostores fail in their first year, 35% within five years, and 46% within ten years (Figure 4.3).

Survival also differs by firm size. Stores with more workers exhibit greater longevity. In the first year, the survival gap between stores with 1–2 workers and those with 6–10 workers

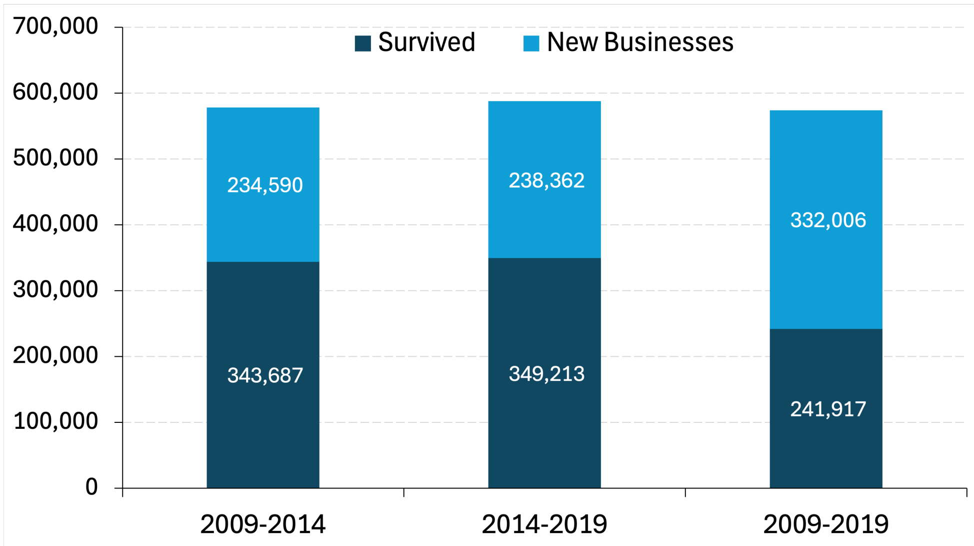

Figure 4.1: New and surviving nanostores across census waves.

is about 5 percentage points. This pattern aligns with the resource-based view of the firm, which posits that access to physical and human capital allows businesses to better withstand operational shocks and maintain continuity in uncertain environments.

4.1 Modeling Determinants of Survival

To explore the determinants of nanostore survival more systematically, we apply two commonly used approaches in duration analysis: the Cox Proportional Hazards (PH) model [71] and the Accelerated Failure Time (AFT) model [72]. Each provides a different lens through which to examine the relationship between covariates and business closure.

4.1.1 Cox Proportional Hazards Model

The Cox Proportional Hazards model, introduced by Cox [71], is a semi-parametric method that models the hazard function $h(t \mid \mathbf{X})$ as a function of time and covariates. The hazard function represents the instantaneous rate of failure at time t, conditional on survival up to that point.

$$h(t \mid \mathbf{X}) = h_0(t) \exp(\boldsymbol{\beta}\mathbf{X})$$

Where: - $h_0(t)$ is the unspecified baseline hazard at time t - $\boldsymbol{\beta}$ is a vector of coefficients - $\mathbf{X}$ is the vector of covariates

The key assumption of the Cox model is proportionality: the hazard ratio between any

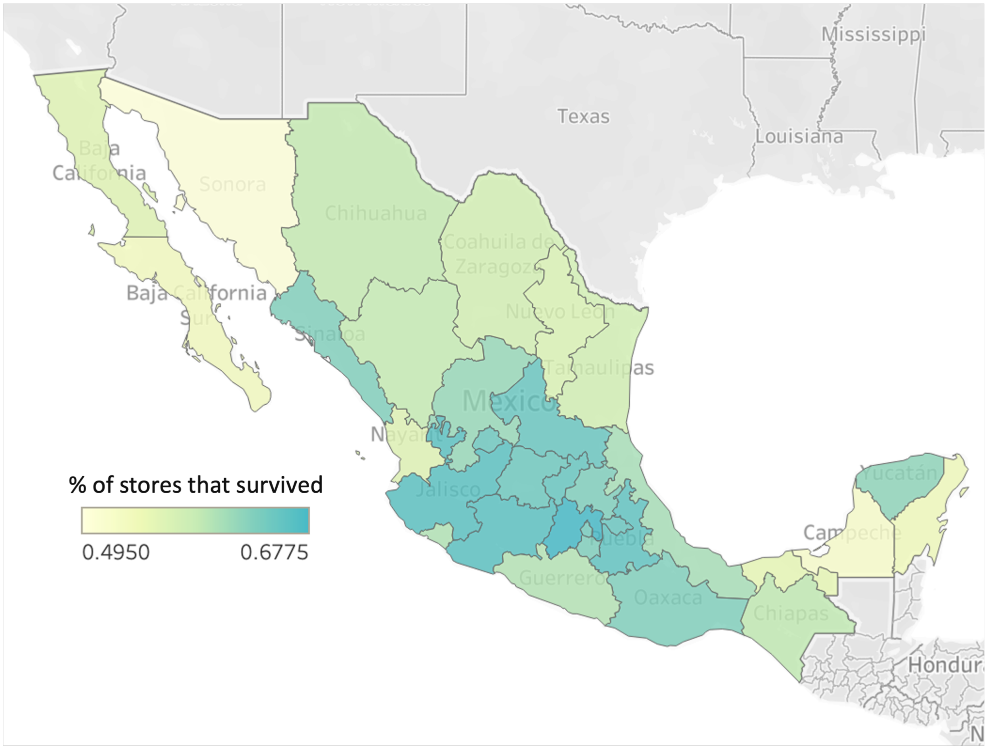

Figure 4.2: Five-year nanostore survival rate by state (2014–2019).

two observations is constant over time and depends only on their covariates. A hazard ratio (HR) greater than 1 implies a higher risk of failure, while an HR less than 1 implies a lower risk. The model does not require specification of the underlying hazard distribution, making it flexible and widely applicable.

4.1.2 Accelerated Failure Time (AFT) Model

The AFT model offers a parametric alternative that focuses directly on the time to failure rather than the hazard rate. It assumes that covariates accelerate or decelerate the entire survival timeline. The logarithm of survival time T is modeled as:

$$\log(T) = \beta_0 + \boldsymbol{\beta}\mathbf{X} + \sigma\epsilon$$

Where: - β_0 is the intercept - $\boldsymbol{\beta}$ is a vector of coefficients - ϵ is a random error term, typically assumed to follow a distribution such as Weibull, exponential, or log-normal - σ is a scale parameter

We use a Weibull distribution for ϵ, as it accommodates increasing or decreasing hazard rates over time—consistent with the high risk of early failure among small firms. In this framework, a negative coefficient indicates that the covariate shortens survival time

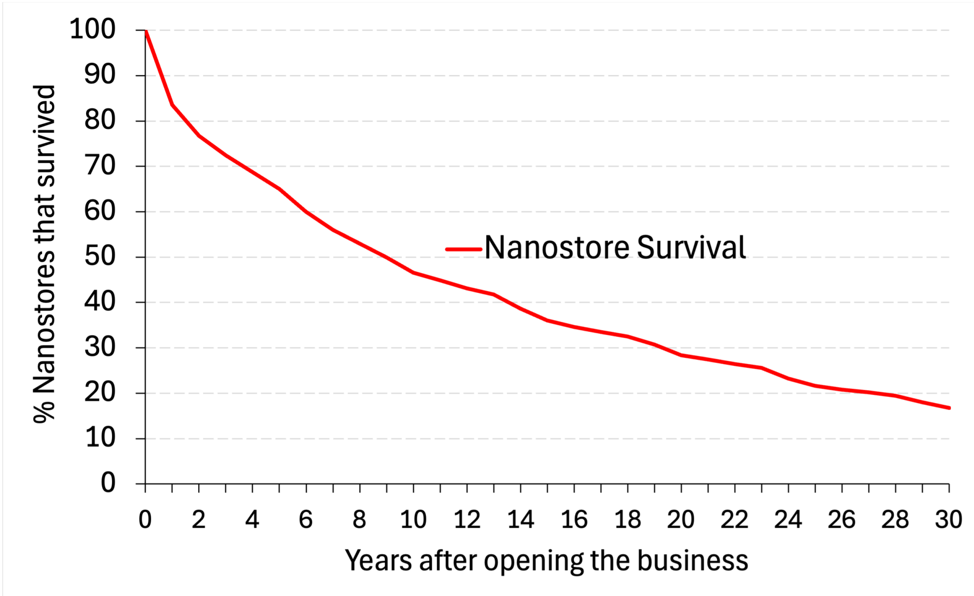

Figure 4.3: Kaplan–Meier survival estimates for nanostores.

(accelerates failure), while a positive coefficient lengthens it (decelerates failure).

4.1.3 Covariates and Interpretation

Both, the Cox proportional and AFT models include the following covariates: the number of male and female workers, average hours worked per employee, the ratio of unpaid workers to total workers, cost-to-income ratio, use of bank cards for business transactions, access to credit, and state fixed effects. These variables capture dimensions of labor, capital structure, financial inclusion, and local regulatory environment.

Table 4.1 presents the estimated coefficients and hazard ratios. For example, in the Cox model, a hazard ratio of 1.16 for bank card usage indicates a 16% higher likelihood of closure at any given time, holding other factors constant. In the AFT model, an exponentiated coefficient of 0.88 for the same variable implies a 12% reduction in expected survival time. The AFT intercept estimate of 1.504 corresponds to a mean predicted survival of approximately 4.5 years.

4.2 Discussion of Survival Rates

The regression results from the Cox Proportional Hazards and Accelerated Failure Time models offer key insights into the determinants of nanostore survival. Specifically, they indicate that firm-level characteristics such as labor composition, financial inclusion (e.g., use

Table 4.1: Cox Proportional Hazards and Accelerated Failure Time Models

Variable	**Cox PH Model (HR)**	**AFT Model (Coef.)**
Number of workers: Men	0.992***	0.006***
Number of workers: Women	0.926***	0.060***
Hours per Worker	0.773***	0.192***
Share of Non Paid Workers	0.364***	0.772***
Ratio Expenses/Income	0.978***	0.016**
Use of Bank Card	1.161***	-0.117***
Use of Credit	0.972***	0.013***
Baja California	0.907***	0.074***
Baja California Sur	0.978	0.016
Campeche	1.079***	-0.061***
Coahuila de Zaragoza	1.059***	-0.047***
Colima	1.085***	-0.067***
Chiapas	1.181***	-0.130***
Chihuahua	0.926***	0.056***
Ciudad de México	0.764***	0.211***
Durango	1.016	-0.008
Guanajuato	0.922***	0.064***
Guerrero	1.000	-0.002
Hidalgo	0.956**	0.040***
Jalisco	0.871***	0.108***
Estado de México	0.932***	0.056***
Michoacán de Ocampo	0.867***	0.112***
Morelos	0.903***	0.077***
Nayarit	0.833***	0.139***
Nuevo León	0.927***	0.057***
Oaxaca	0.767***	0.207***
Puebla	0.929***	0.062***
Querétaro	0.849***	0.125***
Quintana Roo	1.240***	-0.174***
San Luis Potosí	0.929***	0.057***
Sinaloa	0.947***	0.040***
Sonora	0.925***	0.058***
Tabasco	1.135***	-0.101***
Tamaulipas	1.009	-0.007
Tlaxcala	1.013	-0.004
Veracruz de Ignacio de la Llave	0.990	0.009
Yucatán	1.005	-0.002
Zacatecas	0.942***	0.049***

Note: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

A hazard ratio (HR) > 1 implies higher risk of failure; HR < 1 implies longer survival.

In the AFT model, positive coefficients indicate longer time to failure.

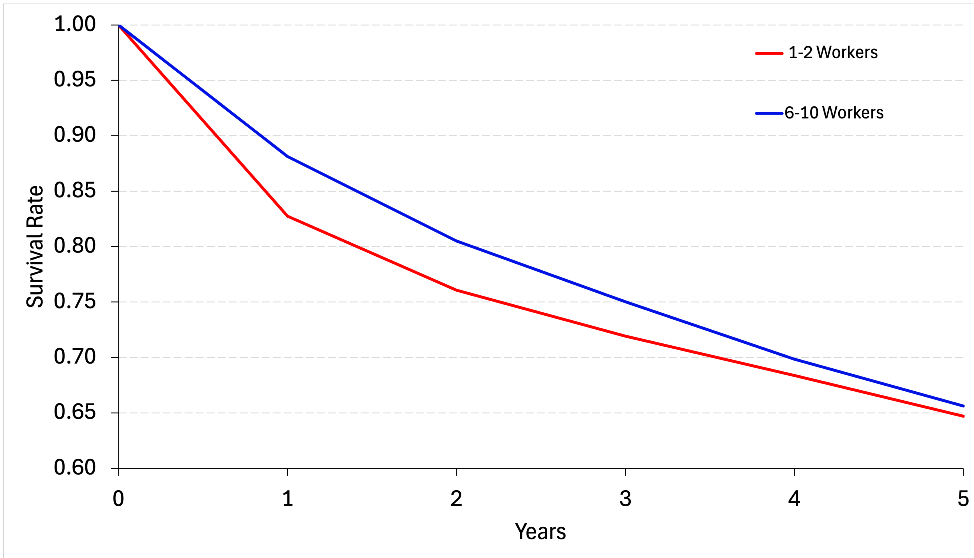

Figure 4.4: Kaplan–Meier survival estimates by firm size.

of credit or digital payment tools), and cost structures significantly influence the likelihood and timing of business closure. For example, stores that rely heavily on unpaid labor or operate with a high cost-to-income ratio face higher hazard rates and shorter expected survival durations. State-level fixed effects further suggest that regulatory and regional business conditions shape survival probabilities across locations. By modeling both the hazard rate and expected failure time, these results provide complementary evidence on how internal and external factors condition the resilience of micro-retail firms in Mexico. This analysis forms a basis for identifying policy-relevant levers to improve firm longevity.

Further extensions could explore interaction effects, time-varying covariates, or heterogeneity by region, firm type, or gender. Together, these survival models provide the empirical foundation for designing targeted interventions to improve the longevity and productivity of Mexico's micro-retail sector.

Chapter 5

Misallocation of Resources

One of the most persistent inefficiencies in developing economies is the misallocation of productive resources across firms. Misallocation occurs when capital and labor are not allocated to the most productive businesses, reducing aggregate output. In the case of nanostores in Mexico —where formal and informal retailers coexist in fragmented and localized markets— distortions in resource allocation constrain firm-level growth and overall sector efficiency.

To analyze this issue, we follow the methodology of Hsieh and Klenow [15], who proposed a framework for measuring productivity losses due to resource misallocation using firm-level data. Their central insight is that, in an efficient economy with no market frictions or institutional barriers, marginal products of inputs should be equalized across firms. Therefore, significant dispersion in marginal revenue products indicates the presence of distortions, such as credit constraints, informality, or poor governance, that prevent efficient reallocation.

5.1 Model and Measurement

Similar to Lagakos [55], production technologies in the micro-retail sector operate by perfectly competitive entrepreneurs, and under a Cobb–Douglas production function:

$$Y_i = A_i K_i^{\alpha} L_i^{(1-\alpha)}$$

Where:

- Y_i: Value Added (VA) of firm i. In our data, this is measured as the value of their total sales minus the value of the goods bought for resale.

- A_i: Total Factor Productivity (TFP). This refers to any means, procedures, or actual technology that helps shopkeepers in improving their sales with the others factors constant. For example, marketing, financial, management, or inventory practices could be considered as technology in this framework given that using the same capital or labor, those practices are considered to improve value added.

- K_i: capital stock. We consider capital stock as the sum of the total assets plus the value of inventory at the beginning of the year.
- L_i: labor input. For this production factor, we consider both formal and informal workers. This, because it is common for this type of businesses to have workers without a formal wage, commonly family members.

Then, the revenue-based measure of total factor productivity (TFPR) is computed as follows:

$$A_i = \frac{Y_i}{K_i^{\alpha} L_i^{(1-\alpha)}}$$

We set the capital share $\alpha = 0.4$, following Lagakos [55]. Value added is preferred over gross revenues to better capture net productivity, particularly in the retail sector where the value created lies in resale margins, inventory management, and operational efficiency. Also, by using the difference between the value of sales and the cost of the goods bought for sales we avoid the problem of considering the total costs of operating the business, which could be misleading due to the fact that not all workers are formally paid.

The marginal revenue products of capital and labor are then defined as:

$$MRPK_i = \alpha \frac{Y_i}{K_i}; \quad MRPL_i = (1-\alpha) \frac{Y_i}{L_i}$$

Firms with marginal revenue products of capital ($MRPK_i$) or labor ($MRPL_i$) significantly above the sample mean are likely input-constrained. These firms generate more output per unit of input and would benefit most from additional resources. In contrast, firms with below-average marginal products may receive more capital or labor than their productivity justifies.

Consider two nanostores: Store A has a high $MRPK$, while Store B has a low $MRPK$. Store A creates more value with each additional unit of capital. As Store A receives more capital, its marginal return eventually declines, approaching the level observed in Store B. This logic suggests that in an efficient allocation, marginal returns equalize across firms.

When $MRPK$ or $MRPL$ varies substantially across firms, the economy exhibits misallocation. Resources flow inefficiently: firms like Store B (with low marginal returns) hold more inputs than they should, while firms like Store A (with high marginal returns) operate below their optimal scale. Reallocating one more unit capital from Store B to Store A raises aggregate output, since Store A produces more value per peso of capital invested.

Therefore, the misallocation of resources means that the resources are not given to the businesses that need it the most and that can add more value to one extra unit of input. We can also refer to store 1 as capital hungry, because even when they can increase their productivity, they face barriers to access to those productivity factors.

The misallocation of inputs could be seen in the spread of the marginal products of inputs in the sector. To quantify distortions, we compute log deviations from the average marginal product:

$$\log(\tau_{K,i}) = \log(MRPK_i) - \log(\overline{MRPK}),$$

$$\log(\tau_{L,i}) = \log(MRPL_i) - \log(\overline{MRPL})$$

Figure 5.1 displays the distributions of marginal products. The variance of $MRPK$ is 2.27—more than twice the 0.93 variance of $MRPL$, suggesting that capital is more misallocated than labor. This finding aligns with the literature showing that financial frictions are a dominant source of inefficiency in developing economies [11, 73, 74].

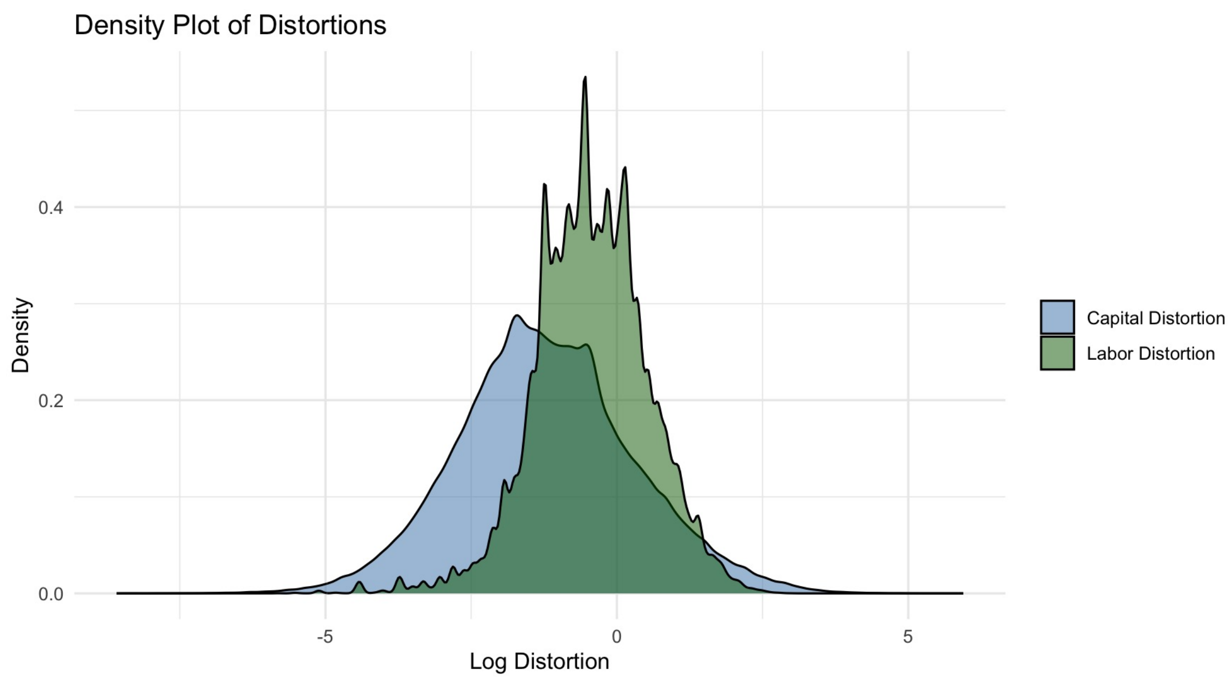

Figure 5.1: Revenue-based marginal product distributions of capital (MRPK) and labor (MRPL).

Furthermore, we observe a negative correlation ($\rho = -0.30$) between TFPR and capital intensity, indicating that less productive firms tend to operate with more capital relative to their output. Therefore, misallocation of resources implies that capital and labor are not flowing to the businesses that could make the most productive use of them, that is, firms capable of generating the highest additional value from one more unit of input. At first glance, this might suggest that low-productivity microenterprises should receive less support, reinforcing a cycle where already-struggling firms remain starved of resources and stagnate. However, the evidence presented here shows that many high-potential nanostores, those with strong fundamentals but limited access to finance or operating in difficult environments, are systematically constrained. This misallocation is not merely a reflection of poor firm quality but often of structural barriers such as informality, financial exclusion, and local crime. In the following sections, we explore in greater depth which factors are associated with improved resource allocation, and how targeted interventions can unlock productivity among constrained but capable firms.

5.2 Regression Evidence

To explore the determinants of misallocation and productivity more systematically, we estimate four regressions using the following dependent variables: value added (VA), total factor productivity (TFPR), marginal product of labor (MRPL), and marginal product of capital (MRPK). Table 5.1 presents the results.

These regressions incorporate both institutional and behavioral variables that may influence firm performance and resource allocation. Institutional factors include elements of the external business environment—such as crime rates, electricity prices, unemployment, urban development, and the Doing Business Index—that shape the constraints and opportunities faced by firms. Behavioral factors, by contrast, refer to internal practices and decisions made by store owners, including their use of bank cards, digital deposits, computational tools, and operational technology, as well as credit source preferences (e.g., bank, supplier, or popular credit). We also include firm age, stock levels, and workforce composition as control variables.

Key Findings:

- **Credit access.**
  - Supplier credit has the largest positive association with firm performance. It increases value added by 20.86, TFPR by 2.176, and MRPL by 9.117, indicating that firms using supplier credit are more productive and make more efficient use of labor. Its effect on MRPK is slightly negative (–0.112) and not statistically significant. This result is in line with Burkart and Ellingsen [75], where the au-

Table 5.1: Regression Analysis

Dependent Variable	(1) VA	(2) TFP	(3) MPL	(4) MPK
Credit from banks	0.473	0.480	1.581	0.045
	(5.852)	(0.701)	(1.467)	(0.169)
Popular credit	13.785**	0.015	2.697*	-0.096
	(6.387)	(0.765)	(1.601)	(0.185)
Credit from suppliers	20.860***	2.176**	9.117***	-0.112
	(7.444)	(0.891)	(1.866)	(0.215)
Credit from government	-8.793	-2.530*	-0.987	-0.435
	(11.450)	(1.371)	(2.870)	(0.331)
Crimes per 1'000 pop	3.858***	0.521***	1.305***	0.048**
	(0.680)	(0.081)	(0.170)	(0.020)
Energy price	0.209	-0.001	0.060	-0.008
	(0.189)	(0.023)	(0.047)	(0.005)
Unemployment	0.656	12.218	50.313*	-1.884
	(108.195)	(12.957)	(27.116)	(3.126)
Doing Business Index	141.487***	8.458*	41.536***	-0.082
	(39.959)	(4.785)	(10.014)	(1.155)
Urban Development Index	143.659***	22.983***	43.483***	3.088***
	(22.346)	(2.676)	(5.600)	(0.646)
Use of technology in op.	43.926***	4.323***	16.634***	-0.105
	(2.511)	(0.301)	(0.629)	(0.073)
Computational assets	1.650***	0.064***	0.201***	-0.001
	(0.080)	(0.010)	(0.020)	(0.002)
Education index	-0.266*	0.065***	0.182***	-0.005
	(0.159)	(0.019)	(0.040)	(0.005)
Age of the business	0.272***	0.016*	0.134***	0.001
	(0.081)	(0.010)	(0.020)	(0.002)
Stores per 1'000 pop.	-1.242	-0.192	-0.235	-0.047
	(1.162)	(0.139)	(0.291)	(0.034)
Total assets ($)	0.054***	-0.020***	0.010***	-0.003***
	(0.003)	(0.0004)	(0.001)	(0.0001)
Stock level ($)	0.857***	-0.00004	0.117***	-0.005***
	(0.012)	(0.001)	(0.003)	(0.0003)
Number of workers	93.676***	-0.231	-9.100***	0.525***
	(2.347)	(0.281)	(0.588)	(0.068)
Convenience stores per 1'000 pop	2.189***	0.405***	0.563***	0.058***
	(0.368)	(0.044)	(0.092)	(0.011)
Constant	-65.125***	22.439***	51.539***	2.321***
	(16.229)	(1.943)	(4.067)	(0.469)
R^2	0.329	0.081	0.091	0.036
Adjusted R^2	0.329	0.080	0.091	0.035
F Statistic	704.575***	126.616***	144.697***	53.354***

Note: Robust standard errors in parentheses; $p^* < 0.1$, $p^{**} < 0.05$, $p^{***} < 0.01$

thors developed a theoretical model to show that it is harder for shopkeepers to deviate in-kind inputs than cash. Moreover, our result supports a better screening for credit that help in reducing the misallocation of resources. Indeed, as suppliers can observe the performance of stores, the asymmetric information is lower than for banks, improving the screening of businesses that can benefit from more capital. Also, is important to notice that trade credit also increases labor productivity. This result does not show evidence of causality. Instead, we notice that this positive correlation could be happening because of workers having more work to do with the products from the trade credit or with suppliers screening which stores have more productive workers. Either of both, this represents a positive impact of trade credit to labor productivity.

  – Bank credit has a modest but significant effect on value added (0.473), but no meaningful effect on TFPR, MRPL, or MRPK. This result is not driven by the small number of nanostores with bank credits because the standard errors are even smaller than the other sources of credit. Instead, this result is showing that banks are not giving credit to the best businesses and/or business owners are not using the credit to increase their productivity.

  – Popular credit is also associated with improved outcomes. It increases value added by 13.785 and MRPL by 2.697, although its effects on TFPR and MRPK are small and statistically insignificant.

  – Government credit, by contrast, has a negative or null effect across all metrics. It reduces TFPR by –2.530 and has a negative but insignificant effect on value added and both marginal products. This suggests either poor targeting or selection bias in how government credit is distributed.

- **Crime.** Higher crime intensity (measured as property crimes per 1,000 residents) is positively and significantly associated with productivity. Value added increases by 3.858, TFPR by 0.521, MRPL by 1.305, and MRPK by 0.048. This is particularly interesting given the general expectation that crime acts as a barrier to productivity. The positive relationship may reflect a selection effect: only the most resilient and productive nanostores survive in high-crime environments. Alternatively, higher crime could indicate densely populated or economically vibrant areas—where both criminal activity and retail opportunity are concentrated. This aligns with findings from Rosenthal and Ross [76], who document that retail establishments tend to cluster in higher-crime areas due to increased foot traffic and commercial density. Thus, while crime imposes direct costs, nanostores operating in such environments may generate higher returns to compensate for the elevated risk.

- **Technology use.**
  - The use of technology in store operations has a strong positive effect on all key productivity indicators: value added increases by 43.93, TFPR by 4.323, and MRPL by 16.63. The effect on MRPK is slightly negative but not statistically significant.
  - Ownership of computational assets (e.g., computers or tablets) is also significant: it increases value added by 1.650, TFPR by 0.064, and MRPL by 0.201.

  The use of technological tools in operations is positively and significantly related to value added, TFPR, and MRPL, suggesting that even basic technologies—such as inventory management systems, electronic point-of-sale tools, or mobile apps—can help nanostores operate more efficiently. Interestingly, there is no significant correlation with MRPK, indicating that while technology improves how labor is deployed, it does not appear to directly ease capital constraints. This may be because the types of technology accessible to nanostores are low-capital (e.g., software or mobile tools), rather than large physical investments. The positive effect is also reinforced by the use of computational assets, which are similarly linked to productivity and labor efficiency.

- **Business environment.**
  - A one-point increase in the Doing Business Index increases value added by 141.49, TFPR by 8.458, and MRPL by 41.536. MRPK remains unaffected.
  - The Urban Development Index shows even stronger effects: value added increases by 143.66, TFPR by 22.983, MRPL by 43.483, and MRPK by 3.088.

  Both the Doing Business Index and the Urban Development Index are positively associated with TFPR and MRPL, and especially with value added. The coefficients are large and significant, suggesting that firms in more efficient and better-governed localities—those with simpler permitting processes, better infrastructure, and stronger institutional frameworks—are substantially more productive. These findings reinforce the importance of localized regulatory and infrastructural reforms. Better local governance may improve access to utilities, reduce bureaucratic delays, and increase the willingness of businesses to formalize, all of which help reduce input distortions and enable better allocation of resources.

- **Human capital.** The education index, which captures the average schooling level of employees in each store, is positively associated with TFPR and MRPL, although its relationship with value added and MRPK is weak or negative. This suggests that better-educated workers may increase the efficiency with which labor is used, but do not necessarily translate into higher firm output unless complementary conditions (such as capital or market access) are also met. This is consistent with prior work showing that education has heterogeneous effects depending on the broader institutional environment [77]. Importantly, this result may also reflect the limits of returns to human capital in low-margin retail environments, where shop tasks are routine, and much of the productivity gain must come from managerial or structural improvements rather than employee education alone.

5.3 Policy Implications

Our analysis reveals significant productivity losses stemming from resource misallocation in the micro-retail sector. The variance of marginal products and negative TFPR–capital correlation imply that more efficient firms are not scaling, while less efficient firms are overusing inputs.

Following the logic of Hsieh and Klenow [15], improving resource allocation across firms—even without innovation or technology adoption—could raise output substantially. Reforms aimed at reducing input distortions, especially in capital markets, can thus unlock latent productivity.

Recommended policy interventions include:

- **Financial inclusion policies** to expand access to productive credit, especially supplier-based and flexible microcredit instruments. Expand supplier-credit schemes and microloan guarantees (For example, Mel, McKenzie, and Woodruff [34] and Banerjee and Duflo [11]). Support fintech platforms for peer-to-peer lending, which have been shown to lower borrowing costs for small retailers [78].

- **Digital adoption incentives** to facilitate technological upgrades that improve inventory control, pricing, and efficiency. Subsidize basic point-of-sale and inventory systems; mobile-money and e-commerce training have measurably increased sales and efficiency in small retail [41, 44].

- **Urban safety and governance reforms** to mitigate the productivity cost of operating in high-crime areas. Improve street lighting and community policing in high-density markets to lower crime-related losses and insurance costs [48, 49].

- **Regulatory simplification** to encourage formality and reduce frictions to entry, investment, and firm growth. Simplify business registration and licensing via one-stop shops (drawing on the "Doing Business" reforms, [47, 50].

Reducing the misallocation of capital and labor across nanostores has the potential to substantially increase the sector's contribution to growth, employment, and poverty reduction.

Chapter 6

MIT LIFT Performance Index

6.1 A Microbusiness Performance Index

Existing data products—such as the INEGI Economic Census, the Monthly Survey on Commercial Establishments (EMEC), and the World Bank's Enterprise Surveys have been invaluable for understanding Mexico's business landscape. They provide nationally representative coverage, a consistent questionnaire over time, and detailed structural variables (revenues, employment, assets). However, these sources were not designed with micro-retailers in mind: sampling frames prioritize larger formal firms, reference periods can be several years apart, and micro-store attrition or informality often causes measurement gaps. Consequently, while they remain indispensable for macro-analysis and long-run trends, they offer limited real-time insight into the day-to-day conditions faced by nanostores.

To address this data vacuum, the MIT Low-Income Firms Transformation Lab[1] is currently working in the LIFT Performance Index, a microbusiness-focused economic indicator designed to provide consistent, and real-time insights into market dynamics at the base of the economic pyramid. Inspired by the Purchasing Managers' Index (PMI), the LIFT Performance Index is a diffusion index that aggregates structured signals from nanostores to track whether business conditions are expanding, contracting, or stable. It allows stakeholders—including suppliers, credit providers, government agencies, and investors—to better understand current conditions and anticipate future trends. Stock and Watson [79] demonstrated how these diffusion indexes are good predictor of future market trends.

The construction of the LIFT Performance Index follows a rigorous, replicable methodology:

1. **Gather stratified sample data:** Each month, we collect data directly from a sample

[1] See webpage https://liftlab.mit.edu

of nanostores stratified by region and sector. Respondents, which are nanostore's shopkeepers, are asked a simple set of standardized questions on sales, customer volume, inventory levels, input prices, and restocking frequency.

2. **Weight observations by store characteristics:** Using population and business density demographics from INEGI, we weight responses to ensure representativeness across different types of businesses and geographic zones.

3. **Weight responses based on factor relevance:** Not all dimensions are equally informative. Sales changes may carry more weight than price fluctuations, for example. We apply factor-specific weights based on past responsiveness and volatility to refine signal extraction.

4. **Adjust for seasonality:** Since nanostore activity is often shaped by holidays, school cycles, and weather patterns, we correct for known seasonal effects to ensure month-to-month comparability.

5. **Disaggregate results by location and sector:** Final index values are split by municipality, state, or sector (e.g., food retail vs. pharmacy vs. services), allowing for nuanced comparisons and localized interventions.

This methodology ensures that the index is not only analytically rigorous but also implementable in real time, using lightweight, replicable survey instruments.

The LIFT Performance Index is more than a measurement tool—it is a solution to the deep information asymmetries that limit access to capital, training, and government support in the nanostore sector. Specifically, the index enables:

- **Credit targeting and expansion:** By identifying high-performing nanostores based on validated operational signals, the index can help suppliers and banks allocate trade credit or microloans more efficiently—reaching firms that are capital-constrained but operationally sound.

- **Smarter government support:** Government agencies can use the index to replace or complement traditional screening mechanisms for subsidy and credit programs, which—as shown in our regression results—have historically underperformed due to poor targeting.

- **Private-sector optimization:** CPG firms and wholesalers can compare company-specific sales trends with local LIFT Performance Index benchmarks to assess market share loss or gain, and adapt distribution strategies based on high-performing areas or sectors.

- **Behavioral diagnostics:** The index can be linked with firm-level characteristics (e.g., tech adoption, gender of owner, use of digital payment systems) to identify what features drive strong performance—enabling targeted business training or incentive schemes.

- **Resilience tracking:** In crisis periods (e.g., post-COVID recovery or inflationary shocks), the index will serve as an early warning system, showing where firms are most vulnerable or resilient, and helping donors and development actors direct emergency support.

One of the key innovations of the LIFT Performance Index is its monthly publication frequency, which contrasts with the 5-year cadence of the Economic Census and the lag of most traditional business surveys. Because it is based on a small set of operational questions and can be digitized (e.g., via SMS or WhatsApp surveys), the index can offer real-time insight into grassroots economic activity. For example, a sudden drop in restocking frequency or customer traffic may signal emerging supply chain issues, insecurity, or local economic downturns.

Construction of the LIFT Performance Index

To address the lack of high-frequency, firm-level data in the micro-retail sector, we construct the LIFT Performance Index, a diffusion-style economic indicator that captures the direction of business conditions across Mexican nanostores. This index is designed to track short-term trends in MSEs activity based on structured survey responses.

The data comes from a nationwide Qualtrics survey, georeferenced and stratified by region, where business owners report on four operational dimensions: (1) number of customers, (2) product variety, (3) inventory levels, and (4) selling prices. These responses are qualitative—classified as *increased*, *unchanged*, or *decreased*—and are transformed into a numerical scoring system to build indicator-specific subindices.

After data cleaning and geolocation, each response is linked to its respective state and municipality using shapefiles and spatial joins. We then use INEGI census data to compute the relative weight of each state in terms of total nanostore count. Each subindex is then calculated as a weighted average of scores by state. The final LIFT Performance Index can be constructed as either an average of these subindices or a custom-weighted composite.

Mathematical Definition

Let:

- $R_{ij} \in \{0, 0.5, 1\}$ denote the re-coded response of firm i for indicator $j \in \{\text{Clientes}, \text{Productos}, \text{Inventario}, \text{Precios}\}$,
- w_s be the weight of state s as a share of total nanostores nationwide,
- $\bar{R}_{sj}$ be the average response for indicator j in state s.

Then the national-level subindex for indicator j is:

$$\text{Index}_j = \sum_s w_s \cdot \bar{R}_{sj}$$

The composite LIFT Performance Index is given by:

$$\text{LIFT Performance Index} = \frac{1}{4}\left(\text{Index}_{Clientes} + \text{Index}_{Productos} + \text{Index}_{Inventario} + \text{Index}_{Precios}\right)$$

Range and Interpretation

Because each qualitative answer is re-coded on the scale $\text{Menos} = 0,\ \text{Igual} = 0.5,\ \text{Más} = 1$, every subindex Index_j is bounded between 0 and 1. For ease of communication we rescale by 100 so that all reported values lie in the closed interval $[0, 100]$.

For the interpretation, a reading of exactly 50 indicates that, on net, the number of businesses reporting expansion equals those reporting contraction (a neutral stance). Values above 50 signal expansion, while values below 50 signal contraction: conditions are worsening for a majority of firms.

Thus, an observed monthly value of 54 would imply a broad-based improvement in micro-retail conditions, whereas a reading of 43 would point to a sector-wide softening. Tracking the index through time provides a simple yet powerful gauge of the direction and momentum of nanostore activity.

Pseudocode for Index Construction

```
1. Load georeferenced Qualtrics survey data (latitude, longitude, responses)
2. Filter invalid entries and responses outside Mexico
3. Recode qualitative answers into numeric values:
     "Mas" → 1, "Igual" → 0.5, "Menos" → 0
```

```
4. Perform spatial join to assign each point to its state and municipality
5. Load INEGI census data with number of nanostores by state
6. For each state:
     a. Compute average response per indicator
     b. Calculate state weight = (# stores in state) / (total national stores)
7. For each indicator:
     Compute national index = sum of (state average × state weight)
8. Combine indicators into composite index:
     LIFT Performance Index = mean of four subindices
9. Export results for visualization or dashboard integration
```

This process enables the creation of a high-frequency, spatially disaggregated economic index that reflects evolving business sentiment and conditions across the micro-retail sector. It also offers a replicable and scalable methodology that can be adapted to future rounds of survey data collection.

For Mexico, with the support from the National Alliance for Small Businesses (ANPEC in spanish), we collected data at the national level. Regarding the performance of the stores, Durango and Mexico City seem to be going through a better time in comparison with other states. On the contrary, Sinaloa and Chihuahua performed poorly. This is the second month in a row with Sinaloa performing poorly (Figure 6.1).

Preliminary results from a pilot implementation of the LIFT Performance Index in Argentina demonstrate both the feasibility and policy relevance of this measurement tool. Conducted in collaboration with the Buenos Aires Institute of Technology (ITBA) and the University of Buenos Aires (UBA), the survey reached over 300 microenterprises across five economic zones and 13 business types. Key findings revealed that more than 50% of businesses had been operating for less than six years, with particularly low survival rates among kiosks and coffee shops. Interviews consistently highlighted inflation, crime, and limited credit access as major operational barriers, especially for sectors like grocery stores and restaurants.

Critically, the index allowed researchers to detect spatial and sectoral differences in performance and to identify emerging trends—such as contraction signals in high-inflation areas or resilience in niche markets like ice cream shops and craft breweries (Figure 6.2). Because many businesses lacked formal financial records, the diffusion-based nature of the LIFT Performance Index proved especially valuable, offering an anonymized, real-time comparison without relying on sensitive or unavailable sales data. These findings underscore the index's value as a diagnostic and decision-making tool for governments, financial institutions, and

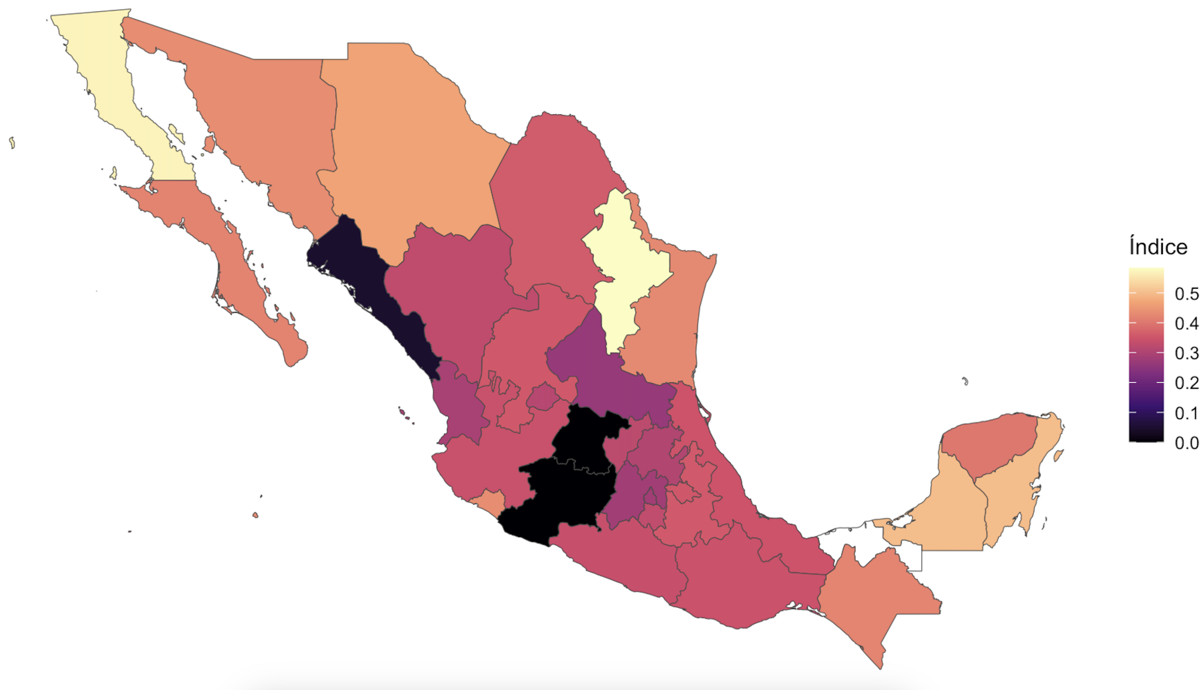

Figure 6.1: Variation of the LIFT Performance Index across different states in Mexico

development organizations seeking to support MSEs in complex macroeconomic environments.

By filling the information gap that separates small businesses from larger economic stakeholders, the LIFT Performance Index creates a common platform through which MSEs can be seen, supported, and scaled. With its low operational burden and high informational value, the index can act as the connective tissue between firms, markets, and public policy—enabling more efficient credit markets, smarter subsidies, better data for CPGs, and more resilient local economies.

The index could play a transformative role in reducing information asymmetries between nanostores and external institutions. Suppliers and trade credit providers could use it to identify trustworthy clients with strong operational habits, even if they lack formal records. Banks could incorporate the index into their alternative credit scoring models, reaching underbanked yet promising firms. Government agencies could target training, infrastructure, or capital programs toward index-verified businesses with demonstrated growth signals.

By functioning as a standardized, transparent measure of microbusiness health, the index would enable more efficient allocation of scarce public and private support. It would also offer policymakers real-time feedback on the conditions facing nanostores, allowing for more responsive and adaptive interventions.

Figure 6.2: Datapoints from Buenos Aires, Argentina. This index can show which areas are growing and which others are contracting.

6.2 Policy Implications

Unlocking the potential of nanostores requires more than just access to capital or regulatory reform. It requires a nuanced, data-informed approach that identifies and supports the right businesses with the right tools: credit, digital infrastructure, business training, and localized policy design, at the right moment in their life cycle. This is especially relevant in environments where formal data systems are weak and institutional support is unevenly distributed.

The proposed LIFT Performance Index contributes to this effort by offering a scalable, real-time diagnostic tool for tracking the conditions and trajectories of micro-retail businesses. By distilling qualitative operational signals into a monthly, diffusion-style index, it addresses one of the core limitations identified throughout this thesis: the lack of high-frequency, reliable data to guide intervention design and credit allocation. If implemented effectively, the index could support more accurate targeting of public programs, foster better-aligned trade credit strategies, and incentivize firms to formalize and grow. Realizing this vision will require collaboration among consumer goods companies, financial institutions, academic researchers, and public agencies, as well as active engagement from the businesses themselves. The potential return through reduced input misallocation, stronger local economies, and more inclusive development is substantial.

Chapter 7

Conclusions

This thesis set out to explain why so many micro-retail businesses—popularly known as nanostores, struggle to survive and grow in Mexico. Although these establishments provide essential goods to more than seventy million consumers and employ over ten percent of the national labor force, they close at alarming rates and exhibit wide productivity gaps. Clarifying the mechanisms behind these outcomes is critical for any policy aimed at inclusive economic development.

The investigation combined four complementary strategies. First, we assembled a longitudinal database by linking the 2009, 2014 and 2019 waves of Mexico's Economic Census, which made it possible to follow more than half a million nanostores over a full decade. Second, we used each firm's record with local information on crime, electricity tariffs, unemployment and regulatory quality so that operating conditions could be taken into account. Third, we applied the misallocation framework developed by Hsieh and Klenow to measure how far capital and labor were from their efficient distribution within the sector. Finally, we designed and piloted the LIFT Performance Index, a diffusion-style indicator that converts monthly survey answers on sales, inventory, prices and staffing into a real-time barometer of micro-retail activity; a first pilot was conducted among more than three hundred microenterprises in Argentina.

7.1 Main Findings and Contributions

The empirical results revealed that only forty-two percent of the nanostores founded in 2009 were still operating ten years later, and the smallest firms failed the fastest. Dispersion in the marginal product of capital was more than twice that of labour, indicating that financial frictions rather than hiring constraints constituted the main brake on productivity. Supplier credit, basic digital tools and a supportive local business environment were strongly associated with higher value added and greater labor efficiency, whereas public credit showed no positive association with any performance metric, suggesting serious targeting problems in government public policies and screening.

The Argentina pilot demonstrated that micro-entrepreneurs, even without formal records, were able and willing to report qualitative changes in their business, and that these signals could be transformed into a timely index that anticipated local market trends. Taken together, the thesis contributed the first nationwide, firm-level portrait of survival, growth and factor misallocation in Mexico's micro-retail sector; it showed that distinct performance outcomes respond to distinct constraints, warning against one-size-fits-all interventions; and it introduced a scalable, low-cost index that can guide credit providers, suppliers and policy-makers toward high-potential firms while respecting data-privacy concerns.

7.2 Limitations

We consider some caveats that accompany the analysis, yet none of them overturn the central results. First, census items such as capital stock are potentially noisy. Because the misallocation exercise relies on relative dispersion, comparing one firm's marginal product to another's, the estimates remain informative so long as measurement error is not systematically correlated with firm-level productivity. Extensive robustness checks using alternative input definitions produced the same qualitative ranking of states and industries, reinforcing confidence in the main patterns.

Second, firms that closed temporarily and later reopened under a new business identifier were counted as exits, which may overstate failure rates. Even under this conservative coding, however, nearly sixty per cent of the firms observed in 2009 disappeared from the census by 2019, a magnitude large enough that any reasonable adjustment for re-entry would still leave Mexico with markedly low survival compared with international benchmarks.

Third, contextual variables on crime, unemployment and utility prices were available only for a single reference year, limiting a full dynamic assessment. Nonetheless, the cross-sectional correlations between these variables and firm performance were strong, statistically precise and consistent with theory, suggesting that the identified relationships capture stable structural forces rather than short-lived shocks.

Finally, the LIFT Performance Index was piloted on a sample of just over three hundred microenterprises in Argentina. While the scale was modest, the pilot convincingly demonstrated both the feasibility of monthly data collection and the index's ability to signal sector expansions and contractions in real time. Replication in additional regions will extend external validity, but the proof-of-concept already shows that a diffusion-style index can solve the information gap that hinders micro-retail policy today.

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